

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\104-D1-Rev 01.pvdb

Date of Analysis : Sep 9, 2026 2:22pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 7196.000 mm OD	
Vessel Design Length, Tangent to Tangent	191.00 cm
Distance of Bottom Tangent above Grade	660.00 cm
Distance of Base above Grade	30.00 cm
Specified Datum Line Distance	630.00 cm
Internal Design Temperature	475 °C
Internal Design Pressure	50.760 kgf/cm ²
External Design Temperature	475 °C
Maximum Allowable Working Pressure	51 kgf/cm ²
Shop Test Pressure	92.653 kgf/cm ²
Field Test Pressure	87.093 kgf/cm ²
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	30.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	
Skirt	SA-516 70	...	...	K02700	
Skirt	SA-387 11	2	...	K11789	
Nozzle	SA-336 F11	2	...	K11572	
Nozzle Flg	SA-182 F11	2	...	K11572	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 ≤ t ≤ 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
sk1	...	...	...	1.6000	N/A	No
SK2	...	...	...	1.6000	N/A	No
SK3	...	...	...	1.6000	N/A	Yes
H1	50.760	0.00	51.000	3.0000	N/A	Yes
S1	50.760	0.00	51.000	3.0000	N/A	Yes
H2	50.760	0.00	51.000	3.0000	N/A	Yes

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-352.5	277.5	38.0	38.0	...	...	1.00	1.00
Skirt	-132.5	220.0	38.0	38.0	...	...	1.00	1.00
Skirt	-0.0	132.5	38.0	38.0	...	...	1.00	1.00



Sphere	-0.0	0.0	92.0	85.0	84.1	...	1.00	1.00
Cylinder	191.0	191.0	168.0	168.0	167.3	...	1.00	1.00
Sphere	191.0	0.0	92.0	85.0	84.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2	WRC-107/537	0.984	n/a	Passed
N1	WRC-107/537	0.984	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	138.75	4496.05	97867.7	27694.1	738863
20	30	387.5	3615.64	97129.8	16439.1	468304
30	40	563.75	2776.42	95674	9407.83	256220
40	50	629.998	2356.04	94550.7	6007.65	130198
50	60	725.5	2356.03	56616.5	6007.58	130196
60	70	820.998	1487.33	30133	2337.21	47351.2

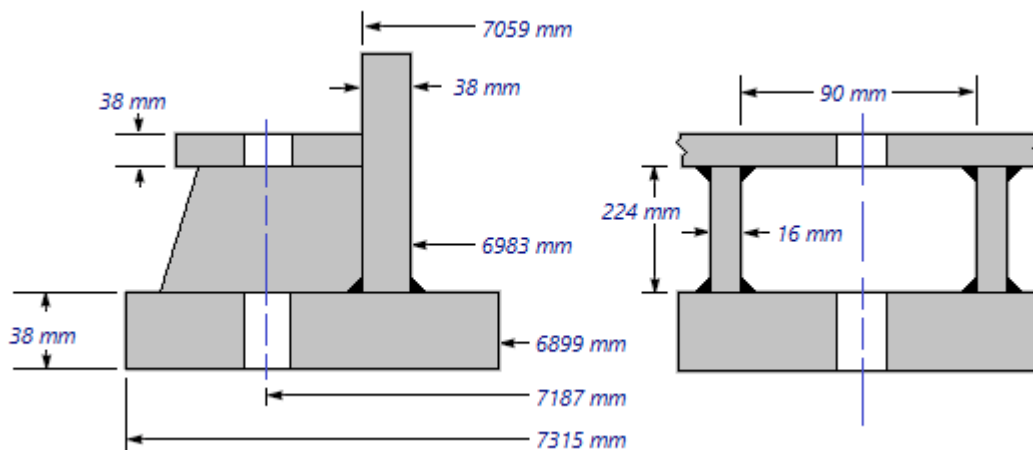
Abs Max of the all of the Stress Ratio's : 0.9683

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	38.0000	mm
Inside Diameter of Basing	6898.9995	mm
Outside Diameter of Basing	7315.0000	mm
Nominal Diameter of Bolts	42.0000	mm
Diameter of Bolt Circle	7187.0000	mm
Number of Bolts	32	

Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	128.0000	mm
Height of Gussets	224.0000	mm
Distance between Gussets	90.0000	mm
Thickness of Top Plate or Ring	38.0000	mm
Radial Width of the Top Plate	128.0000	mm
Number of Gussets	64	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View



Loads for Foundation/Support Design:

Total Wind Shear on Support	13773. kgf
Total Earthquake Shear on Support	107145. kgf
Wind Moment on Support	170513. kgf-m.
Earthquake Moment on Support	881682. kgf-m.

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	261599.4 kg
Shop Test - Fabricated + water (full)	506721.7 kg
Shipping - Fab. + rem. intls.+ shipping app.	419155.9 kg
Erected - Fab. + rem. intls.+ insul. (etc)	419155.9 kg
Empty - Fab. + intls. + details + wghts.	419155.9 kg
Operating - Empty + operating liquid (no ca)	624155.9 kg
Field Test - Empty weight + test medium (full)	533560.5 kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2	51	...	...	OK	25.45	OK	Passed
UC1	51	...	...	...		OK	Passed
H1	51	...	...	...		OK	Passed
K4	51	...	...	...		OK	Passed
K2	51	...	...	...		OK	Passed
N1	51	...	...	OK	25.45	OK	Passed
M1	51	...	...	...		OK	Passed
K1	51	...	...	...		OK	Passed
K3	51	...	...	OK	6.42	OK	Passed
K5	51	...	...	...		OK	Passed
K6	51	...	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 51 Nozzle : K2
 Minimum MAWP Shells/Flanges : 51 Element : S1
 Minimum MAPnc Shells/Flanges : 72 Element : S1

Computed Vessel M.A.W.P. : 51 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
40	N2	0.000	0.000	1840.800
40	UC1	0.000	0.000	1231.200
40	H1	0.000	0.000	383.075
40	K4	0.000	180.000	252.900
50	K2	955.000	0.000	390.325
60	N1	0.000	0.000	1840.800
60	M1	0.000	180.000	1231.200
60	K1	0.000	0.000	224.325
60	K3	0.000	180.000	224.325
60	K5	0.000	0.000	278.300
60	K6	0.000	180.000	278.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2	2.000 in	XXS	WNF	2.375	11.074	...	...	518.94	600
K1	2.000 in	XXS	WNF	2.375	11.074	...	...	963.40	600
K3	2.000 in	XXS	WNF	2.375	11.074	...	...	936.48	600
K4	3.000 in	XXS	WNF	3.500	15.240	...	...	388.98	600
K5	4.000 in	XXS	WNF	4.500	17.120	...	...	1177.68	600
K6	4.000 in	XXS	WNF	4.500	17.120	...	...	2573.66	600
H1	8.000 in	160	WNF	8.625	23.012	...	...	475.50	600
UC1	24.000 in	Actual	WNF	25.575	20.000	...	...	2435.00	600
M1	24.000 in	Actual	WNF	25.575	20.000	...	...	1204.42	600
N2	36.000 in	Actual	WNF	38.205	28.000	...	...	657.88	600
N1	36.000 in	Actual	WNF	38.205	28.000	...	...	751.77	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2	SA-336 F11	84.000	10.000	...	...	...
K1	SA-336 F11	84.000	10.000	...	...	...
K3	SA-336 F11	84.000	10.000	...	...	...
K4	SA-336 F11	84.000	10.000	...	...	...
K5	SA-336 F11	84.000	10.000	...	...	...
K6	SA-336 F11	84.000	10.000	...	...	...
H1	SA-336 F11	84.000	10.000	...	...	...
UC1	SA-336 F11	84.000	10.000	...	...	14.000
M1	SA-336 F11	84.000	10.000	...	...	...
N2	SA-336 F11	84.000	10.000	...	...	...
N1	SA-336 F11	84.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2	95.497	0.0	350.00	0.00	S1
K1	...	0.0	850.00	0.00	H2
K3	...	180.0	826.85	0.00	H2
K4	...	180.0	267.45	0.00	H1
K5	...	0.0	1026.85	0.00	H2
K6	...	180.0	2200.00	0.00	H2
H1	...	0.0	317.45	0.00	H1
UC1	...	0.0	1850.00	500.00	H1
M1	...	180.0	750.00	0.00	H2
N2	...	0.0	500.00	0.00	H1
N1	...	0.0	600.00	0.00	H2



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 92.0mm THK X 6935.0mm ID X 0.0cm	SA-387 11
3	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 4000.0mm	...
1	PACKING: 610.000mm	...
2	INSULATION: 3430mm X 150mm THK	HOT INSULATION
1	LADDER: 2325.0mm	...
1	PACKING: 1910.000mm	...
1	INSULATION: 1910mm X 150mm THK	HOT INSULATION
2	LADDER: 2700.0mm	...
1	PLATFORM: 1000mm X 5000mm WIDE	
1	PACKING: 620.000mm	...
2	CLASS 600 GR 1.9, 24.0" BLIND FLANGE(S)	SA-182 F11
1	CLASS 600 GR 1.9, 8.0" BLIND FLANGE(S)	SA-182 F11
1	CLASS 600 GR 1.9, 3.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 2.0" BLIND FLANGE(S)	SA-182 F11
2	CLASS 600 GR 1.9, 4.0" BLIND FLANGE(S)	SA-182 F11
1	BASERING: 7315.0mm OD X 6899.0 ID X 38.0mm	SA-36
32	BASERING BOLTS: 42.0mm DIA	SA-307 A
32	BASERING BOLT NUTS: 42.0mm DIA	...
1	BASERING TOP RING: 7315.0mm OD X 7059.0 ID X 38.0mm	...
64	BASERING GUSSET PLATES: 128.0mm X 128.0mm X 224.0mm X 16.0mm	...
5	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3107.8mm X 277.5cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3046.9mm X 277.5cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1584.4mm X 277.5cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 20170.2mm X 277.5cm	SA-516 70
5	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3107.8mm X 220.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3046.9mm X 220.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1584.4mm X 220.0cm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 20170.2mm X 220.0cm	SA-516 70
6	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 3102.6mm X 132.5cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 1551.3mm X 132.5cm	SA-387 11
1	CYLINDRICAL SKIRT SEGMENT 38.0mm THK X 20166.9mm X 132.5cm	SA-387 11
2	CYLINDRICAL SEGMENT 168.0mm THK X 8560.5mm X 191.0cm	SA-387 11

PV Elite 28 U2 Licensee: SL User
FileName : 104-D1-Rev 01
Bill of Material:

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HEXAGON

1	CYLINDRICAL SEGMENT 168.0mm THK X 4429.9mm X 191.0cm	SA-387 11
1	NAMEPLATE	...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
H1	[9]	A	46	30		0.718	85.000	1.00	Yes
S1	[8]	!	-12	-27		0.727	168.000	1.00	Yes
H2	[9]	A	46	30		0.718	85.000	1.00	Yes
N2	[1]	A	46	30		0.714	85.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
UC1	[1]	A	46	30		0.714	85.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
H1	[1]	A	-7	-104		0.199	20.136	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K4	[1]	A	-20	-104		0.122	13.335	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K2	[1]	B	-29	-104		0.129	9.690	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
N1	[1]	A	46	30		0.714	85.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
M1	[1]	A	46	30		0.714	85.000	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K1	[1]	A	-29	-104		0.129	9.690	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K3	[1]	A	-29	-104		0.129	9.690	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K5	[1]	A	-16	-104		0.139	14.980	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
K6	[1]	A	-16	-104		0.139	14.980	1.00	Yes
Nozzle Flg	[5]	B	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 46 30

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature 30.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:



Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	50.76	kgf/cm ²
Design Internal Temperature	475.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		30	cm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	cm
Distance Upwind of Crest	Lh	0	cm
Distance from Crest to the Vessel	x	0	cm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	cm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		2.000	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		2.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Design Pressure + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	Yes
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	sk1	
Distance "FROM" to "TO"	277.5	cm
Skirt Inside Diameter	6983	mm
Diameter of Skirt at Base	6983	mm
Skirt Thickness	38	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	50	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2606.3	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	tailing lug	
Dist. from "FROM" Node / Offset dist	-0.0004069	cm
Miscellaneous Weight	2500	kgf
Offset from Element Centerline	3525	mm

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	SK2	
Distance "FROM" to "TO"	220	cm
Skirt Inside Diameter	6983	mm
Diameter of Skirt at Base	6983	mm
Skirt Thickness	38	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	50	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Platform	
Detail ID	PF1	
Dist. from "FROM" Node / Offset dist	150	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	15000	cm ²
Platform Weight	1937.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	185	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	30.0	
Ladder Start Elevation	0	cm
Ladder End Elevation	400	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	SK3	
Distance "FROM" to "TO"	132.5	cm
Skirt Inside Diameter	6983	mm
Diameter of Skirt at Base	6983	mm
Skirt Thickness	38	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	475	°C
Design Temperature External Pressure	475	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	1509.1	kgf/cm ²
Allowable Stress, Operating	1091.6	kgf/cm ²
Allowable Stress, Hydrotest	2844.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	407.48	kgf/cm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Head-to-Skirt or Circ. Seam	1.0	
Weld is pre-Heated	No	

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Spherical	
Description	H1	
Distance "FROM" to "TO"	0	cm
Inside Diameter	6935	mm
Element Thickness	85	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.76	kgf/cm ²
Design Temperature Internal Pressure	475	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	475	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Packing	
Detail ID	CATALYST	
Dist. from "FROM" Node / Offset dist	-62	cm
Height of Packed Section	61	cm
Density	0.00112	kg/cm ³
Percent Volume Holdup	0.0	



Specific Gravity of Packing Liquid	0.0	
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	-343	cm
Height/Length of Insulation	343	cm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	UC1	
Dist. from "FROM" Node / Offset dist	2550	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	H1	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	8	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	237.66	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	3	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	46.011	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	
Element From Node	40	
Detail Type	Weight	
Detail ID	Internal DE	
Dist. from "FROM" Node / Offset dist	-343	cm
Miscellaneous Weight	4460	kgf
Offset from Element Centerline	0	mm
Element From Node	40	



Detail Type	Weight	
Detail ID	Wire Cloth	
Dist. from "FROM" Node / Offset dist	-263	cm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Inert ballast b	
Dist. from "FROM" Node / Offset dist	-150	cm
Miscellaneous Weight	157000	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Davit-h1	
Dist. from "FROM" Node / Offset dist	-300	cm
Miscellaneous Weight	50	kgf
Offset from Element Centerline	1200	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	wire screen	
Dist. from "FROM" Node / Offset dist	-60	cm
Miscellaneous Weight	100	kgf
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	SF1	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	7110	kgf
Offset from Element Centerline	0	mm

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	S1	
Distance "FROM" to "TO"	191	cm
Element Outside Diameter	7196	mm
Element Thickness	168	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	168	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.76	kgf/cm ²
Design Temperature Internal Pressure	475	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	475	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PF2	
Dist. from "FROM" Node / Offset dist	0	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	30000	cm ²
Platform Weight	7192.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	45.0	
Ladder Start Elevation	400	cm
Ladder End Elevation	632.5	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm

Element From Node	50	
Detail Type	Packing	
Detail ID	CATALYST	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	191	cm
Density	0.00112	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	191	cm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K2	
Dist. from "FROM" Node / Offset dist	95.5	cm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	54.836	kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	

Element From Node	50	
Detail Type	Weight	
Detail ID	CLIP-WT	
Dist. from "FROM" Node / Offset dist	95.5	cm
Miscellaneous Weight	4045	kgf
Offset from Element Centerline	0	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	misc margin	
Dist. from "FROM" Node / Offset dist	100	cm
Miscellaneous Weight	10000	kgf
Offset from Element Centerline	0	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Spherical	
Description	H2	
Distance "FROM" to "TO"	0	cm
Inside Diameter	6935	mm
Element Thickness	85	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	50.76	kgf/cm ²
Design Temperature Internal Pressure	475	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	475	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	



Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Platform	
Detail ID	PF-3	
Dist. from "FROM" Node / Offset dist	79	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	0	cm ²
Platform Weight	1878.2	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	0.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	632.5	cm
Ladder End Elevation	902.5	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	0	mm
Element From Node	60	
Detail Type	Platform	
Detail ID	PF-4	
Dist. from "FROM" Node / Offset dist	368	cm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	50000	cm ²
Platform Weight	4352	kgf
Platform Railing Weight	0	kgf/cm
Platform Grating Weight	0.017	kgf/cm ²
Platform Width	5000	mm
Platform Height	1000	mm
Platform Clearance or End Offset	2500	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	632.5	cm
Ladder End Elevation	902.5	cm
Unit Weight of Ladder	0.37	kgf/cm
Platform Length (top head platform)	5000	mm
Element From Node	60	
Detail Type	Packing	
Detail ID	catalyst	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	62	cm
Density	0.00112	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	343	cm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm ³
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N1	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0) 0 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60
Detail Type Nozzle
Detail ID M1
Dist. from "FROM" Node / Offset dist 1800 mm
Nozzle Diameter 24 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 180.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 0 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60
Detail Type Nozzle
Detail ID K1
Dist. from "FROM" Node / Offset dist 1200 mm
Nozzle Diameter 2 in.
Nozzle Schedule XXS
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 30.899 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60
Detail Type Nozzle
Detail ID K3
Dist. from "FROM" Node / Offset dist 1075 mm
Nozzle Diameter 2 in.
Nozzle Schedule XXS
Nozzle Class 600
Layout Angle 180.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 26.424 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60
Detail Type Nozzle
Detail ID K5
Dist. from "FROM" Node / Offset dist 1600 mm
Nozzle Diameter 4 in.
Nozzle Schedule XXS
Nozzle Class 600
Layout Angle 0.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 96.271 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60
Detail Type Nozzle
Detail ID K6
Dist. from "FROM" Node / Offset dist 3000 mm
Nozzle Diameter 4 in.
Nozzle Schedule XXS
Nozzle Class 600
Layout Angle 180.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 0 kgf
Grade of Attached Flange GR 1.9
Nozzle Matl SA-336 F11

Element From Node 60



Detail Type	Weight	
Detail ID	trunnion1	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	2500	kgf
Offset from Element Centerline	3430	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	trunnion2	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	2500	kgf
Offset from Element Centerline	3430	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Davit	
Dist. from "FROM" Node / Offset dist	300	cm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	1800	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Wire Screen	
Dist. from "FROM" Node / Offset dist	60	cm
Miscellaneous Weight	100	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	Bed support bal	
Dist. from "FROM" Node / Offset dist	67.5	cm
Miscellaneous Weight	48000	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	gas wiper ring	
Dist. from "FROM" Node / Offset dist	120	cm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	SF2	
Dist. from "FROM" Node / Offset dist	0	cm
Miscellaneous Weight	7110	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	Weight	
Detail ID	internal pipe	
Dist. from "FROM" Node / Offset dist	340	cm
Miscellaneous Weight	100	kgf
Offset from Element Centerline	0	mm
Element From Node	60	
Detail Type	For./Mom.	
Detail ID	N1	
Dist. from "FROM" Node / Offset dist	343	cm
Force in X Direction	6560	kgf
Force in Y Direction	-9083	kgf
Force in Z Direction	6560	kgf
Moment about X Axis	18165	kgf-m.
Moment about Y Axis	0	kgf-m.
Moment about Z Axis	18165	kgf-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
	sk1	...	277.5	...	277.5
	SK2	...	497.5	...	220
	SK3	...	630	...	132.5
	H1	...	630	...	...
	S1	...	821	...	191
	H2	...	821	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
	sk1	...	38	1.6	6983	...	1
	SK2	...	38	1.6	6983	...	2
	SK3	...	38	1.6	6983	...	3
	H1	50.76	92	3	6935	1091.6	4
	S1	50.76	168	3	7196	1091.6	5
	H2	50.76	92	3	6935	1091.6	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
	sk1	...	No Calc	No Calc	38	No Calc	1
	SK2	...	No Calc	No Calc	38	No Calc	2
	SK3	...	No Calc	No Calc	38	No Calc	3
	H1	50.76	51.3422	73.6252	85	84.0658	4
	S1	50.76	50.9958	71.8049	168	167.251	5
	H2	50.76	51.3422	73.6252	85	84.0658	6

Minimum

51

72

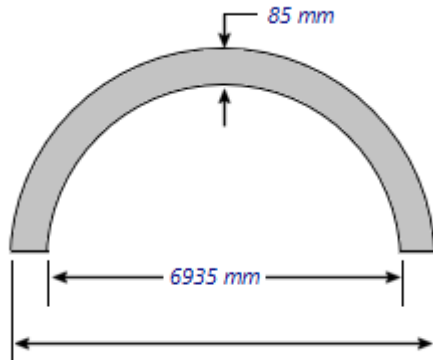
MAWP: 51 kgf/cm², limited by: S1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: H1

Spherical Head Design Information:					
Design Pressure	P	50.760 kgf/cm²	Int. Design Temperature	475 °C	
Material		SA-387 11	Ext. Design Temperature	475 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1091.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	6935.000 mm	Actual Stress	1079.2 kgf/cm²	
Spec. Min. Thk t _e , or s _a		85.0000 mm	Spec. Nominal Thk.	92.000 mm	
Surface Area		796082.1 cm²	SG of Contents	0.000	
Element Volume		87318856.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		55308.8 kgf	Operating Weight	55308.8 kgf	
			Comp. All. Ope. Stress	677.9 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 86.688 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 43.344 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.76 \cdot 3470.4998) / (2 \cdot 1091.62 \cdot 1.0 - 0.2 \cdot 50.76)$$

$$= 81.0658 + 3.0000 = 84.0658 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1091.62 \cdot 1.0 \cdot 82.0) / (3470.4998 + 0.2 \cdot 82.0)$$

$$= 51 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 85.0) / (3467.5 + 0.2 \cdot 85.0)$$

$$= 74 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.76 \cdot (3470.4998 + 0.2 \cdot 82.0)) / (2 \cdot 1.0 \cdot 82.0)$$

= 1079.238 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.964 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 85.0$, $t_r = 58.837$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.718$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 46 °C
Min Metal Temp. at Required thickness (UCS 66.1) 30 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	50.996 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	85.000 mm	Basic MDMT	46 °C
Stress/Thk. Ratio	0.718	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	30 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

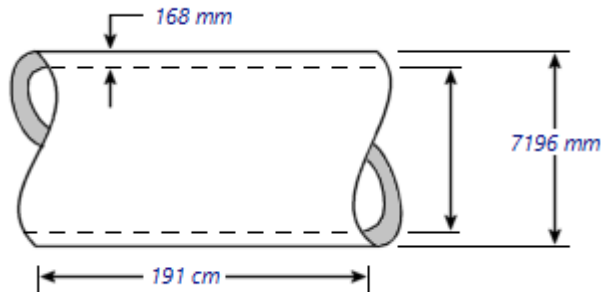
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 86.688 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 43.344 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S1

Cylindrical Shell Design Information:					
Design Pressure	P	50.760 kgf/cm²	Int. Design Temperature	475 °C	
Material		SA-387 11	Ext. Design Temperature	475 °C	
Length		191.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1091.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Outside Diameter	Do	7196.000 mm	Actual Stress	1086.6 kgf/cm²	
Spec. Min. Thk t _e , or s _a		168.0000 mm	Spec. Nominal Thk.	168.000 mm	
Surface Area		431791.8 cm²	SG of Contents	0.000	
Element Volume		70594600.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		54909.4 kgf	Operating Weight	54909.4 kgf	
			Comp. All. Ope. Stress	767.6 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (50.76 \cdot 3598.0) / (1091.62 \cdot 1.0 + 0.4 \cdot 50.76)$$

$$= 164.2509 + 3.0000 = 167.2509 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1091.62 \cdot 1.0 \cdot 165.0) / (3598.0 - 0.4 \cdot 165.0)$$

$$= 51 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1509.1 \cdot 1.0 \cdot 168.0) / (3598.0 - 0.4 \cdot 168.0)$$

$$= 72 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$$

$$= (50.76 \cdot (3598.0 - 0.4 \cdot 165.0)) / (1.0 \cdot 165.0)$$

$$= 1086.572 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / Ro))$ 2.390 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

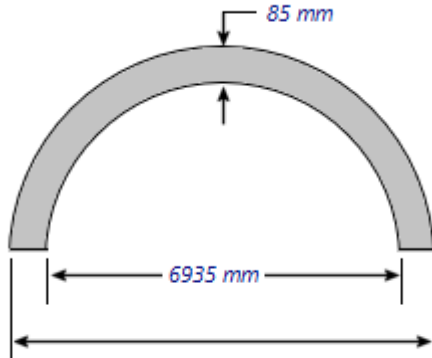
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: H2

Spherical Head Design Information:					
Design Pressure	P	50.760 kgf/cm²	Int. Design Temperature		475 °C
Material		SA-387 11	Ext. Design Temperature		475 °C
Length		0.000 cm	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1091.6 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1509.1 kgf/cm²
Inside Diameter	Di	6935.000 mm	Actual Stress		1079.2 kgf/cm²
Spec. Min. Thk t _e , or sa		85.0000 mm	Spec. Nominal Thk.		92.000 mm
Surface Area		796082.1 cm²	SG of Contents		0.000
Element Volume		87318856.0 cm³	Weight of Contents		0.0 kgf
Empty Weight		55308.8 kgf	Operating Weight		55308.8 kgf
			Comp. All. Ope. Stress		677.9 kgf/cm²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 86.688 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 43.344 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (50.76 \cdot 3470.4998) / (2 \cdot 1091.62 \cdot 1.0 - 0.2 \cdot 50.76)$$

$$= 81.0658 + 3.0000 = 84.0658 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1091.62 \cdot 1.0 \cdot 82.0) / (3470.4998 + 0.2 \cdot 82.0)$$

$$= 51 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 1509.1 \cdot 1.0 \cdot 85.0) / (3467.5 + 0.2 \cdot 85.0)$$

$$= 74 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (50.76 \cdot (3470.4998 + 0.2 \cdot 82.0)) / (2 \cdot 1.0 \cdot 82.0)$$



= 1079.238 kgf/cm²

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.964 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 85.0$, $t_r = 58.837$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.718$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 46 °C
Min Metal Temp. at Required thickness (UCS 66.1) 30 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	50.996 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	85.000 mm	Basic MDMT	46 °C
Stress/Thk. Ratio	0.718	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	30 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 86.688 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 43.344 mm

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	87.093 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	86.690 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	92.653 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	73.694 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	83.356 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	86.690 kgf/cm ²
App 27-4	= M.A.W.P.	50.996 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
H1	1984.7	2844.9	0.698	93.35
S1	1998.7	2844.9	0.703	93.37
H2	1984.7	2844.9	0.698	93.35

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2	Nozzle	1407.00	1071.00	1.314
UC1	Nozzle	1407.00	1071.00	1.314
H1	Nozzle	1407.00	1071.00	1.314
K4	Nozzle	1407.00	1071.00	1.314
K2	Nozzle	1407.00	1071.00	1.314
N1	Nozzle	1407.00	1071.00	1.314
M1	Nozzle	1407.00	1071.00	1.314
K1	Nozzle	1407.00	1071.00	1.314
K3	Nozzle	1407.00	1071.00	1.314
K5	Nozzle	1407.00	1071.00	1.314
K6	Nozzle	1407.00	1071.00	1.314

Minimum 1.314

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H1	1509.10	1091.62	1.382
S1	1509.10	1091.62	1.382
H2	1509.10	1091.62	1.382

Minimum 1.382

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2	1774.33	2524.00	0.703
UC1	1746.13	2524.00	0.692
H1	559.36	2524.00	0.222
K4	364.14	2524.00	0.144
K2	383.62	2524.00	0.152
N1	1774.33	2524.00	0.703
M1	1746.13	2524.00	0.692
K1	383.52	2524.00	0.152
K3	383.52	2524.00	0.152
K5	407.98	2524.00	0.162
K6	407.98	2524.00	0.162



Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	87.093 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	86.690 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	92.809 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	73.694 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	83.356 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	86.690 kgf/cm ²
App 27-4	= M.A.W.P.	50.996 kgf/cm ²

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 50.996 • 1.314
 = 87.093 kgf/cm²

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
H1	1870.5	2844.9	0.658	87.98
S1	1875.8	2844.9	0.659	87.63
H2	1859.1	2844.9	0.653	87.44

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2	Nozzle	1407.00	1071.00	1.314
UC1	Nozzle	1407.00	1071.00	1.314
H1	Nozzle	1407.00	1071.00	1.314
K4	Nozzle	1407.00	1071.00	1.314
K2	Nozzle	1407.00	1071.00	1.314
N1	Nozzle	1407.00	1071.00	1.314
M1	Nozzle	1407.00	1071.00	1.314
K1	Nozzle	1407.00	1071.00	1.314
K3	Nozzle	1407.00	1071.00	1.314
K5	Nozzle	1407.00	1071.00	1.314
K6	Nozzle	1407.00	1071.00	1.314

Minimum 1.314

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
H1	1509.10	1091.62	1.382
S1	1509.10	1091.62	1.382
H2	1509.10	1091.62	1.382

Minimum 1.382

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N2	1672.27	2524.00	0.663
UC1	1645.69	2524.00	0.652
H1	527.18	2524.00	0.209
K4	343.19	2524.00	0.136
K2	360.03	2524.00	0.143
N1	1662.05	2524.00	0.658
M1	1635.63	2524.00	0.648



K1	359.25	2524.00	0.142
K3	359.25	2524.00	0.142
K5	382.16	2524.00	0.151
K6	382.16	2524.00	0.151

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	630	...	...	No Calc	No Calc
20	30	630	...	...	No Calc	No Calc
30	40	630	...	...	No Calc	No Calc
40	50	No Calc	7105	82	No Calc	No Calc
50	60	422.167	7196	165	No Calc	No Calc
60	70	No Calc	7105	82	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	85	No Calc	...	11.2844
50	60	168	No Calc	...	59.7507
60	70	85	No Calc	...	37.2519

Minimum

11

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	630	No Calc	No Calc	No Calc
20	30	630	No Calc	No Calc	No Calc
30	40	630	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	422.167	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	20473.2	...	19714.2	...	...
20	30	14291.5	...	13689.8	...	...
30	40	8607.34	...	8244.99	...	...
40	50	55308.8	87318856	53550.8	87545680	...
50	60	54909.4	70594600	53951.9	70718136	...
60	70	55308.8	87318856	53550.8	87545680	...
Total		208898	0.245E+09	202702	0.246E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Wght	2500	-352.5	-0.00040691	...	tailing lug
20	Plat	1937.16	284.496	150	-284.496	PF1
40	Pack	25556.5	...	-31.5	...	CATALYST
40	Insl	4586.57	...	-171.5	...	Ins: 20
40	Nozl	2809.93	...	-367.5	...	N2
40	Nozl	2693.62	255	-298.22	...	UC1
40	Nozl	237.664	120	-336.946	...	H1
40	Nozl	46.0107	-120	-586.43	...	K4
40	Wght	4460	...	-343	...	Internal DE
40	Wght	150	...	-263	...	Wire Cloth
40	Wght	157000	...	-150	...	Inert ballast b
40	Wght	50	-120	-300	...	Davit-h1
40	Wght	100	...	-60	...	wire screen
40	Wght	7110	...	...	...	SF1
50	Plat	7192.16	0.38155E-05	...	...	PF2
50	Pack	79157.1	...	95.5	...	CATALYST
50	Insl	2330.07	...	95.5	...	Ins: 20
50	Nozl	54.836	360.501	95.5	...	K2
50	Wght	4045	...	95.5	...	CLIP-WT
50	Wght	10000	...	100	...	misc margin
60	Plat	1878.24	279.831	79	-279.831	PF-3
60	Plat	4352.01	...	368	...	PF-4
60	Pack	25980.1	...	31	...	catalyst
60	Insl	4586.57	...	171.5	...	Ins: 20
60	Nozl	3268.61	...	372.5	...	N1
60	Nozl	2200.25	-180	329.621	...	M1
60	Nozl	30.8994	120	363.574	...	K1
60	Nozl	26.4235	-107.5	366.758	...	K3
60	Nozl	96.2705	160	354.721	...	K5
60	Nozl	161.06	-300	279.634	...	K6
60	Wght	2500	...	...	-343	trunnion1
60	Wght	2500	...	...	343	trunnion2
60	Wght	150	-180	300	...	Davit
60	Wght	100	...	60	...	Wire Screen
60	Wght	48000	...	67.5	...	Bed support bal
60	Wght	200	...	120	...	gas wiper ring
60	Wght	7110	...	...	...	SF2
60	Wght	100	...	340	...	internal pipe
60	Forc	...	...	343	...	N1

Total Weight of Each Detail Type:

Platforms	15359.6
Packing	130693.7
Insulation	11503.2
Nozzles	11625.6
Weights	246075.0

Sum of the Detail Weights 415257.0 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	208898.9	208898.9	208898.9	208898.9	208898.9	208898.9
Nozzles	11625.6	11625.6	11625.6	11625.6	11625.6	11625.6
Wld Weights	41075.0	41075.0	41075.0	41075.0	41075.0	41075.0
Platforms	...	...	15359.6	15359.6	15359.6	15359.6
Packing	...	...	130693.7	130693.7	130693.7	130693.7
Insulation	...	...	11503.2	11503.2	11503.2	11503.2
Ope Weights	...	...	...	...	...	205000.0
Test Liquid	...	245122.2	...	...	...	...
Totals	261599.4	506721.7	419155.9	419155.9	419155.9	624155.9

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	261599.4 kg
Shop Test Wt.	- Fabricated weight + water (full)	506721.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	419155.9 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	419155.9 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	419155.9 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	624155.9 kg
Field Test Wt.	- Empty weight + test medium (full)	533560.5 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	411876.2 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	608320.3 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	368575.7 kg

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	615398	608772
20	30	487883	482630
30	40	293839	290675
40	50	796082	755462
50	60	431792	411630
60	70	796082	755462

Total 3421075.5 3304633.0 cm²
 Total 3682.4 3557.1 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	22973.2	22973.2	22214.2	...	22973.2
20	30	16228.6	16228.6	15626.9	5511.15	16228.6
30	40	8607.34	8607.34	8244.99	...	8607.34
40	50	103109	260109	166076	...	260109
50	60	157689	157689	148260	...	157689
60	70	110549	158549	173138	5255.9	158549

Empty Support Force + the Sum of the Y forces 428238.91 kgf
 Operating Support Force + the Sum of the Y forces 633238.94 kgf



Hydro Support Force + the Sum of the Y forces 542643.50 kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	624156	624156	533560
20	30	601183	601183	511346
30	40	584954	584954	495719
40	50	576347	576347	487474
50	60	316238	316238	321399
60	70	158549	158549	173138

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	10767	10767	10767
20	30	10767	10767	10767
30	40	5255.9	5255.9	5255.9
40	50	5255.9	5255.9	5255.9
50	60	5255.9	5255.9	5255.9
60	70	5255.9	5255.9	5255.9

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2	70.16	105.44	475	600	GR 1.9	...	...	...	...
UC1	70.16	105.44	475	600	GR 1.9	...	...	...	...
H1	70.16	105.44	475	600	GR 1.9	...	...	...	...
K4	70.16	105.44	475	600	GR 1.9	...	...	...	...
K2	70.16	105.44	475	600	GR 1.9	...	...	...	...
N1	70.16	105.44	475	600	GR 1.9	...	...	...	...
M1	70.16	105.44	475	600	GR 1.9	...	...	...	...
K1	70.16	105.44	475	600	GR 1.9	...	...	...	...
K3	70.16	105.44	475	600	GR 1.9	...	...	...	...
K5	70.16	105.44	475	600	GR 1.9	...	...	...	...
K6	70.16	105.44	475	600	GR 1.9	...	...	...	...
Min Rating	70.159	105.441, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 11.2319 Hz.

The Natural Frequency for the Vessel (Ope...) is 11.2319 Hz.

The Natural Frequency for the Vessel (Filled) is 12.1716 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	9277.24	57510.2

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. kgf	X,Z Moment and For Res kgf-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	9277.24	57510.2

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
60	343.00	6560.	-9083.	6560.	18165.		18165.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	30 cm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.225 \cdot 0.928) / (1 + 1.7 \cdot 3.4 \cdot 0.225)))$
 $= \min(0.85, 0.887)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	11.232 Hz
Natural Frequency of Vessel (Empty)	11.232 Hz
Natural Frequency of Vessel (Test)	12.172 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.513
Structure Height to Diameter ratio	1.767
Height to top of Structure	1191.250 cm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 27432.0$ cm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 138.75 + 30.0 = 168.75$ cm
 $= 5.536$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (5.536 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 27432.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.119 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.513 \cdot 253.022$$

$$= 1764.5 \text{ lbs. } [800.4 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
sk1	168.8	0.000	0.000	0.000	0.849	1.000	78.119
SK2	417.5	0.000	0.000	0.000	0.849	1.000	78.119
SK3	593.7	0.000	0.000	0.000	0.897	1.000	78.119
H1	660.0	0.000	0.000	0.000	0.917	1.000	78.119
S1	755.5	0.000	0.000	0.000	0.944	1.000	78.119
H2	1008.1	0.000	0.000	0.000	1.003	1.000	78.119

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/821.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[23.49/26.94^2] \cdot 10^4 = 323.804$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.0004$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 617960 / (821.0 \cdot 721.58^2)$$

$$= 0.14456E-02$$

Since Vp is > 0.0004 no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf	Cf
PF1	15000.00	457.50	78.12	117.18	1.00
PF2	30000.00	660.00	78.12	234.36	1.00
PF-3	0.00	930.00	78.12	0.00	0.00
PF-4	50000.00	1219.00	78.12	390.59	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
tailing lug	0.00	30.00	78.12	0.00
Internal DE	0.00	317.00	78.12	0.00
Wire Cloth	0.00	397.00	78.12	0.00
Inert ballast	0.00	510.00	78.12	0.00
Davit-h1	0.00	360.00	78.12	0.00
wire screen	0.00	600.00	78.12	0.00
SF1	0.00	660.00	78.12	0.00
CLIP-WT	0.00	755.50	78.12	0.00
misc margin	0.00	760.00	78.12	0.00
trunnion1	0.00	851.00	78.12	0.00
trunnion2	0.00	851.00	78.12	0.00
Davit	0.00	1151.00	78.12	0.00
Wire Screen	0.00	911.00	78.12	0.00
Bed support b	0.00	918.50	78.12	0.00
gas wiper rin	0.00	971.00	78.12	0.00
SF2	0.00	851.00	78.12	0.00
internal pipe	0.00	1191.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 11.2319 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	168.75	847.08	235065	78.1189	880.409
20	30	417.5	847.08	186358	78.1189	839.227
30	40	593.75	847.08	112238	78.1189	420.376
40	50	659.998	888.6	2.70845	78.1189	0.010144
50	60	755.5	899.52	171808	78.1189	868.695
60	70	1008.14	888.6	258402	78.1189	1487.33

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(38.3120.75)$
= 0.308 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/11.232)
= 0.089

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.308$, 0.089) per 12.8.2
= 0.089

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 2.0)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.089(2.0 / 2.0))$
= 1.168

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 2.0), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1568 \cdot 624156$
= 97867.648 kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 624156$



= 19573.5 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 11.2319 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	138.75	22973.2	737.888	736.492	10490.5
20	30	387.5	16228.6	1455.76	1453.01	9770.09
30	40	563.75	8607.34	1123.29	1121.17	9261.16
40	50	629.998	260109	37934.3	15009	8991.24
50	60	725.5	157689	26483.4	26433.3	834.215
60	70	820.998	158549	30133	20970.7	-4110.9



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support cm	Cumulative Shr Wind Cas kgf	Cumulative Shr Eqk Cas kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	138.75	9277.24	9277.24	133676	133676
20	30	387.5	9277.24	9277.24	107932	107932
30	40	563.75	9277.24	9277.24	87521.8	87521.8
40	50	629.998	9277.24	9277.24	75229.7	75229.7
50	60	725.5	9277.24	9277.24	75229.4	75229.4
60	70	820.998	9277.24	9277.24	57510.2	57510.2

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	138.75	4496.05	97867.7	27694.1	738863
20	30	387.5	3615.64	97129.8	16439.1	468304
30	40	563.75	2776.42	95674	9407.83	256220
40	50	629.998	2356.04	94550.7	6007.65	130198
50	60	725.5	2356.03	56616.5	6007.58	130196
60	70	820.998	1487.33	30133	2337.21	47351.2

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	9277.24	0.0011973	0.0046305	0.32228E-05
20	30	9277.24	0.0087836	0.014073	0.52898E-05
30	40	9277.24	0.017757	0.021782	0.63193E-05
40	50	9277.24	0.021782	0.021782	0.63193E-05
50	60	9277.24	0.027887	0.034123	0.65909E-05
60	70	9277.24	0.034123	0.034123	0.65909E-05

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 41.050 Actual deflection : 0.034 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	4496.05	0.00023834	0.00088178	0.58529E-06
20	30	3615.64	0.001614	0.0024949	0.85631E-06
30	40	2776.42	0.0030841	0.0037111	0.97098E-06
40	50	2356.04	0.0037111	0.0037111	0.97098E-06
50	60	2356.03	0.0046446	0.0055872	0.99051E-06
60	70	1487.33	0.0055872	0.0055872	0.99051E-06

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	1707.96	10674.8
20	30	1707.96	10674.8
30	40	1707.96	10674.8
40	50	1791.68	11198
50	60	1813.69	11335.6
60	70	1791.68	11198

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 41.050 Actual deflection : 0.006 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 41.050 Actual deflection : 0.040 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	8381.69	8028.82	1463323904	1402028416
20	30	8381.69	8028.82	1463323904	1402028416
30	40	8381.69	8028.82	1463323904	1402028416
40	50	18745.8	18092	3.251E+09	3.140E+09
50	60	37092.9	36446.1	6.369E+09	6.263E+09
60	70	18745.8	18092	3.251E+09	3.140E+09

Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1688.56	2863.21	-1134.68	-1134.68
20	30	1688.56	2863.21	-1134.68	-1134.68
30	40	1309.94	3413.83	-563.197	-1361.78
40	50	1309.94	3413.83	-677.858	-1422.05
50	60	1309.94	3413.83	-767.569	-1473.42
60	70	1309.94	3413.83	-677.858	-1422.05

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	1064.01	...	1825.6
50	60	517.906	...	888.611
60	70	1064.01	...	1825.6

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	-77.7395	-77.7395	-66.4557	0.76796	0.76796
20	30	-74.8782	-74.8782	-63.6889	0.76796	0.76796
30	40	-72.8569	-72.8569	-61.7426	0.37488	0.37488
40	50	-31.8565	-31.8565	-31.8565	0.16737	0.16737
50	60	-8.67687	-8.67687	-8.67687	0.083922	0.083922
60	70	-8.76351	-8.76351	-8.76351	0.16737	0.16737

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	0.76796	1.97529	52.6996	0.98764	...
20	30	0.76796	1.17252	33.4019	0.58626	...
30	40	0.37488	0.67102	18.2749	0.33551	...
40	50	0.16737	0.19131	4.14609	0.095655	...
50	60	0.083922	0.095924	2.07886	0.047962	...
60	70	0.16737	0.074427	1.50787	0.037213	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	35.9783
20	30	...	...	...	23.0426
30	40	...	...	...	12.9591
40	50	...	...	...	3.12879
50	60	...	...	...	1.56878
60	70	...	...	...	1.04938

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
------	----	---	--



10	20	-1.1313	1.30661
20	30	-1.1313	1.21688
30	40	-1.1313	1.15349
40	50	-0.50205	0.49697
50	60	-0.24922	0.022889
60	70	-0.50205	-0.22722

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	9.53448	9.53448	9.1351	9.1351
20	30	7.69826	7.69826	7.37579	7.37579
30	40	6.24251	6.24251	5.98103	5.98103
40	50	2.39564	2.39564	2.31404	2.31404
50	60	1.2012	1.2012	1.18124	1.18124
60	70	1.83138	1.83138	1.76899	1.76899

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-91.15	No Calc		0.0497
20		1688.56	-85.65	No Calc		0.0479
30		1309.94	-81.28	No Calc		0.2981
40		1309.94	-35.11	No Calc		0.0000
50		1309.94	-10.31	No Calc		0.0057
60		1309.94	-11.34	No Calc		0.0000

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-122.71	No Calc		0.0481
20		1688.56	-105.17	No Calc		0.0464
30		1309.94	-91.28	No Calc		0.2889
40		1309.94	-37.05	No Calc		0.0000
50		1309.94	-11.51	No Calc		0.0055
60		1309.94	-12.04	No Calc		0.0000

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-91.15	No Calc		0.0497
20		1688.56	-85.65	No Calc		0.0479



30	1309.94	-81.28	No Calc	0.2981
40	1309.94	-35.11	No Calc	0.0000
50	1309.94	-10.31	No Calc	0.0057
60	1309.94	-11.34	No Calc	0.0000

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-139.43	No Calc		0.0481
20		1688.56	-115.53	No Calc		0.0464
30		1309.94	-96.60	No Calc		0.2889
40		1309.94	-38.07	No Calc		0.0000
50		1309.94	-12.02	No Calc		0.0055
60		1309.94	-12.50	No Calc		0.0000

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		2863.21	-68.21	No Calc		0.0419
20		2863.21	-65.04	No Calc		0.0401
30		3413.83	-62.45	No Calc		0.2488
40		3413.83	-32.12	No Calc		0.0000
50		3413.83	-8.81	No Calc		0.0055
60		3413.83	-8.97	No Calc		0.0000

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		2863.21	-67.22	No Calc		0.0419
20		2863.21	-64.46	No Calc		0.0401
30		3413.83	-62.12	No Calc		0.2488
40		3413.83	-32.02	No Calc		0.0000
50		3413.83	-8.76	No Calc		0.0055
60		3413.83	-8.93	No Calc		0.0000

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-91.15	No Calc		0.0497
20		1688.56	-85.65	No Calc		0.0479
30		1309.94	-81.28	No Calc		0.2981
40	1034.41	1309.94		No Calc	0.7897	
50	510.36	1309.94		No Calc	0.3896	
60	1056.82	1309.94		No Calc	0.8068	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-139.43	No Calc		0.0481
20		1688.56	-115.53	No Calc		0.0464
30		1309.94	-96.60	No Calc		0.2889
40	1039.36	1309.94		No Calc	0.7934	
50	512.62	1309.94		No Calc	0.3913	
60	1058.53	1309.94		No Calc	0.8081	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-91.15	No Calc		0.0497
20		1688.56	-85.65	No Calc		0.0479
30		1309.94	-81.28	No Calc		0.2981
40		1309.94	-35.11	No Calc		0.0000
50		1309.94	-10.31	No Calc		0.0057
60		1309.94	-11.34	No Calc		0.0000



Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-139.43	No Calc		0.0481
20		1688.56	-115.53	No Calc		0.0464
30		1309.94	-96.60	No Calc		0.2889
40		1309.94	-38.07	No Calc		0.0000
50		1309.94	-12.02	No Calc		0.0055
60		1309.94	-12.50	No Calc		0.0000

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		2863.21	-68.21	No Calc		0.0419
20		2863.21	-65.04	No Calc		0.0401
30		3413.83	-62.45	No Calc		0.2488
40	1794.01	3413.83		No Calc	0.5255	
50	880.07	3413.83		No Calc	0.2578	
60	1817.05	3413.83		No Calc	0.5323	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		2863.21	-67.22	No Calc		0.0419
20		2863.21	-64.46	No Calc		0.0401
30		3413.83	-62.12	No Calc		0.2488
40	1793.92	3413.83		No Calc	0.5255	
50	880.02	3413.83		No Calc	0.2578	
60	1817.01	3413.83		No Calc	0.5322	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-78.51	No Calc		0.0490
20		1688.56	-75.65	No Calc		0.0472
30		1309.94	-73.23	No Calc		0.2936
40	1032.32	1309.94		No Calc	0.7881	
50	509.31	1309.94		No Calc	0.3888	
60	1055.41	1309.94		No Calc	0.8057	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-74.47	No Calc		0.0469
20		1688.56	-71.73	No Calc		0.0452
30		1309.94	-69.79	No Calc		0.2812
40	1033.26	1309.94		No Calc	0.7888	
50	509.38	1309.94		No Calc	0.3889	
60	1055.55	1309.94		No Calc	0.8058	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-78.51	No Calc		0.0490
20		1688.56	-75.65	No Calc		0.0472
30		1309.94	-73.23	No Calc		0.2936
40	1032.32	1309.94		No Calc	0.7881	
50	509.31	1309.94		No Calc	0.3888	
60	1055.41	1309.94		No Calc	0.8057	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
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10		1688.56	-78.51	No Calc	0.0490
20		1688.56	-75.65	No Calc	0.0472
30		1309.94	-73.23	No Calc	0.2936
40	1032.32	1309.94		No Calc	0.7881
50	509.31	1309.94		No Calc	0.3888
60	1055.41	1309.94		No Calc	0.8057

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-78.51	No Calc		0.0490
20		1688.56	-75.65	No Calc		0.0472
30		1309.94	-73.23	No Calc		0.2936
40		1309.94	-32.02	No Calc		0.0000
50		1309.94	-8.76	No Calc		0.0055
60		1309.94	-8.93	No Calc		0.0000

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1407.13	-86.74	No Calc		0.0578
20		1407.13	-82.13	No Calc		0.0557
30		1091.62	-78.32	No Calc		0.3467
40	1035.21	1091.62		No Calc	0.9483	
50	510.54	1091.62		No Calc	0.4677	
60	1057.02	1091.62		No Calc	0.9683	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1407.13	-86.74	No Calc		0.0578
20		1407.13	-82.13	No Calc		0.0557
30		1091.62	-78.32	No Calc		0.3467
40		1091.62	-33.92	No Calc		0.0000
50		1091.62	-9.94	No Calc		0.0066
60		1091.62	-10.99	No Calc		0.0000

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Unity Check
10		1688.56	-78.51	No Calc		0.0490
20		1688.56	-75.65	No Calc		0.0472
30		1309.94	-73.23	No Calc		0.2936
40	1032.32	1309.94		No Calc	0.7881	
50	509.31	1309.94		No Calc	0.3888	
60	1055.41	1309.94		No Calc	0.8057	

Absolute Maximum of the all of the Stress Ratio's 0.9683

Governing Element: H2

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
sk1	...	-76.4	0.8	9.5	...	+ Tens
sk1	...	-76.4	-0.8	-9.5	-86.7	- Comp
SK2	...	-73.7	0.8	7.7	...	+ Tens
SK2	...	-73.7	-0.8	-7.7	-82.1	- Comp
SK3	...	-71.7	0.4	6.2	...	+ Tens
SK3	...	-71.7	-0.4	-6.2	-78.3	- Comp



H1	1064.0	-31.4	0.2	2.4	1035.2	+ Tens
H1	1064.0	-31.4	-0.2	-2.4	...	- Comp
S1	517.9	-8.7	0.1	1.2	510.5	+ Tens
S1	517.9	-8.7	-0.1	-1.2	...	- Comp
H2	1064.0	-9.0	0.2	1.8	1057.0	+ Tens
H2	1064.0	-9.0	-0.2	-1.8	...	- Comp

Additional Results for Code Case 2286-1

Nomenclature :

- L - Section Length cm
 Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , kgf/cm²
 Fba - Allowable all comp mem stress due to bending moment, kgf/cm²
 Fva - Allowable shear stress when only shear is present, kgf/cm²
 Fha - Allowable hoop comp memb stress under ext pressure alone, kgf/cm²
 Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , kgf/cm²
 Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , kgf/cm²
 Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , kgf/cm²
 Fbha - Allowable axial comp from bending w/hoop compression, kgf/cm²
 Fhba - Allowable hoop comp w/long compression due to moment, kgf/cm²

Allowable Stress Results for Case 1 : NP+EW+WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 3 : NP+OW+WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.029	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.071	4221	1302	1651	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 6 : NP+HW+HE



From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.029	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.071	4221	1302	1651	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 7 : IP+OW+WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 9 : EP+OW+WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.029	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.071	4221	1302	1651	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.029	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.071	4221	1302	1651	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307



Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
20	0.074	6300	1322	1322	711		70307	1322	70307	70307	70307
30	0.032	6300	206	206	124		70307	206	70307	70307	70307
40							70307		70307	70307	70307
50	0.078	4221	1302	1619	664	1302	70307	1302	70307	70307	70307
60							70307		70307	70307	70307



Skirt Support Analysis Per EN 13445-3 Section 16.12:

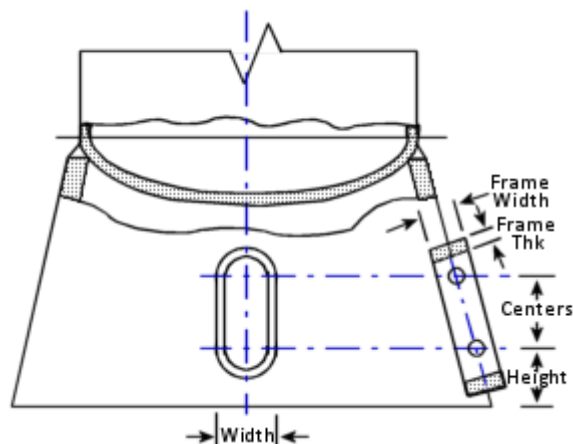
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: sk1

Total vessel (empty) weight	Fg	580783.875 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	580783.875 kgf
Skirt Mean Diameter	Dz	7021.000 mm
Skirt Analysis Thickness (Corroded)	ez	36.400 mm
Skirt Height	H	2775.000 mm
Additional Axial Force at section 1-1	F1	-9083.000 kgf
Moment at section 1-1	M1	576235.812 kgf-m.
Applied Moment - Bottom of Skirt	Mb	872539.250 kgf-m.
Weight at Bottom of Skirt	W	624155.938 kgf



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	1000.0000	90.0	1395.00	0.00	28.000	200.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	81.8115	98.1885

The points for integrating lxx, lyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-261.812	81.812



Height of Weakest Section	H	1394.172	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-0.000	mm
Centroid Position in Y direction	h	-114.288	mm
Principle Moment of Inertia	Ixx	462519296.000	cm4
Principle Moment of Inertia	Iyy	494595808.000	cm4
Distance to Extreme Fiber	ybar	3642.988	mm
Distance to Extreme Fiber	xbar	3528.700	mm
Minimum Section Modulus	W4	1269616384.000	mm3
Metal Area at the Cutting Plane	A4	7775.552	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(1000.0/7021.0)$$

$$= 0.1429 \text{ rad } (8.1885 \text{ deg})$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 7021.0/2$$

$$= 3510.5 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.143 \cdot \sqrt{(3510.5/36.4)}$$

$$= 1.404$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [e_z]:

$$= e_x(\text{new}) - c_i - c_{ext}$$

$$= 38.0 - 0.8 - 0.8$$

$$= 36.4 \text{ mm}$$

Find the moment acting at the Critical Height [M_4]:

$$= (M_{bottom} - M_{top}) \cdot (H - \text{CutHeight}) / H + M_{top}$$

$$= (872539 - 576236) \cdot (2775.0 - 1394.172) / 2775.0 + 576236$$

$$= 723675 \text{ kgf-m.}$$

Compute the section modulus [Z_{xx} and Z_{yy}]:

$$Z_{xx} = I_{xx} / ybar = 0.46252E+09 / 3642.988 = 0.12696E+10 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / xbar = 0.49460E+09 / 3528.7 = 0.14016E+10 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(0.12696E+10, 0.14016E+10) = 0.12696E+10 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))}$$

$$= \sqrt{(\min(0.46252E+09, 0.49460E+09) / 7775.552)}$$

$$= 2438.931 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 2438.931 / 2775.0$$

$$= 0.879$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(-0.0^2 + -114.288^2)}$$

$$= 88.864 \text{ mm}$$

$$\text{Where } e_x = -0.0 : e_y = -114.288$$

Moment Increment at Centroid [ΔM_4]:



$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-9083.0 - 580784 - 0.0) \cdot 114.288 \\
 &= 67414.828 \text{ kgf-m.}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(28.0 - 36.4) / 2 \\
 &= 4.200 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 1000.0 - 200.0 \\
 &= 800.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(800.0 / 7021.0) \\
 &= 0.114 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 28.0 \cdot 200.0 \\
 &= 112.000 \text{ cm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.114) \cdot 7021.0 \cdot 36.4 \\
 &= 7736.959 \text{ cm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 7736.959 + 112.0 \\
 &= 7848.959 \text{ cm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 7021.0 \cdot \cos(0.114) + 4.2 - 0.5 \cdot 28.0 \\
 &= 3477.837 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\
 &= (0.5 \cdot 7021.0 \cdot 36.4 \cdot 800.0 - 2 \cdot 28.0 \cdot 200.0 \cdot 3477.837) / 7848.959 \\
 &= 80.615 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 7021.0 \cdot \cos(0.114) + 4.2 + 80.615); \text{abs}(0.5 \cdot 7021.0 - 80.615)] \\
 &= 3572.451 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.114 - \sin(0.114) \cdot \cos(0.114)) \cdot 36.4 \cdot (7021.0/2)^3 \\
 &= 458909568.0 \text{ cm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (0.45891E+09 + 112.0(3477.837^2 + 28.0^2/12) - 7848.959 \cdot 80.615^2) / 3572.451 \\
 &= 1321072768.000 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 7848.959 / (\pi \cdot 7021.0 \cdot 36.4)) \\
 &= 0.978
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2/3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 0.13211E+10 / (\pi \cdot 7021.0^2 \cdot 36.4)) \\
 &= 0.937
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 7021.0^2 \cdot 36.4 \cdot 1134.7 \\ &= 15990470.0 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 7021.0 \cdot 36.4 \cdot 1134.7 \\ &= 9110071.0 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(yq)) / (\psi_{i2} \cdot M_{max}) \\ &= 589867 / (0.978 \cdot 9110071) + (\text{abs}(723675) + 589867 \cdot \text{abs}(80.615)) / \\ &\quad (0.937 \cdot 15990470) \\ &= 0.118 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

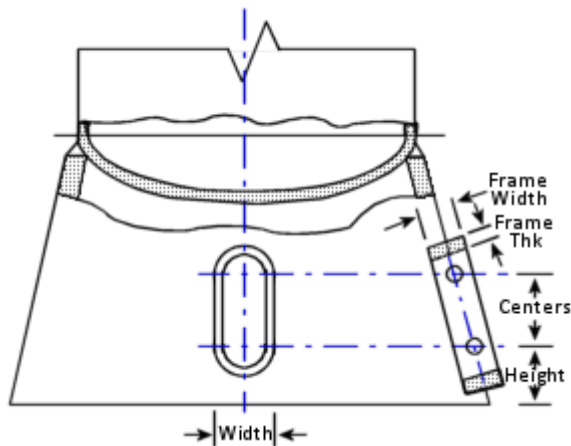
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: SK2

Total vessel (empty) weight	Fg	580783.875 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	580783.875 kgf
Skirt Mean Diameter	Dz	7021.000 mm
Skirt Analysis Thickness (Corroded)	ez	36.400 mm
Skirt Height	H	2200.000 mm
Additional Axial Force at section 1-1	F1	-9083.000 kgf
Moment at section 1-1	M1	343741.406 kgf-m.
Applied Moment - Bottom of Skirt	Mb	576235.812 kgf-m.
Weight at Bottom of Skirt	W	601182.750 kgf



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	850.0001	0.0	700.00	0.00	200.000	28.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-6.9536	6.9536

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-353.046	-6.954



Height of Weakest Section	H	699.370	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	-88.864	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	494720320.000	cm4
Principle Moment of Inertia	Iyy	469667712.000	cm4
Distance to Extreme Fiber	ybar	3528.700	mm
Distance to Extreme Fiber	xbar	3617.564	mm
Minimum Section Modulus	W4	1298299264.000	mm3
Metal Area at the Cutting Plane	A4	7830.633	cm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / \text{Dz})$$

$$= \sin^{-1}(850.0/7021.0)$$

$$= 0.1214 \text{ rad (6.9536 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= \text{Dz} / 2$$

$$= 7021.0/2$$

$$= 3510.5 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.121 \cdot \sqrt{(3510.5/36.4)}$$

$$= 1.192$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= e_{x(\text{new})} - c_i - c_{\text{ext}}$$

$$= 38.0 - 0.8 - 0.8$$

$$= 36.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$$

$$= (576236 - 343741.406) \cdot (2200.0 - 699.37) / 2200.0 + 343741.406$$

$$= 502327 \text{ kgf-m.}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{\text{bar}} = 0.49472\text{E}+09 / 3528.7 = 0.14020\text{E}+10 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{\text{bar}} = 0.46967\text{E}+09 / 3617.564 = 0.12983\text{E}+10 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(0.14020\text{E}+10, 0.12983\text{E}+10) = 0.12983\text{E}+10 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A4))}$$

$$= \sqrt{(\min(0.49472\text{E}+09, 0.46967\text{E}+09) / 7830.633)}$$

$$= 2449.047 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 2449.047 / 2200.0$$

$$= 1.113$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(-88.864^2 + 0.0^2)}$$

$$= 114.288 \text{ mm}$$

$$\text{Where } e_x = -88.864 : e_y = 0.0$$

Moment Increment at Centroid [$\Delta M4$]:



$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-9083.0 - 580784 - 0.0) \cdot 88.864 \\
 &= 52418.184 \text{ kgf-m.}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(200.0 - 36.4) / 2 \\
 &= 81.800 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 850.0 - 28.0 \\
 &= 822.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(822.0 / 7021.0) \\
 &= 0.117 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 200.0 \cdot 28.0 \\
 &= 112.000 \text{ cm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.117) \cdot 7021.0 \cdot 36.4 \\
 &= 7728.897 \text{ cm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 7728.897 + 112.0 \\
 &= 7840.897 \text{ cm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 7021.0 \cdot \cos(0.117) + 81.8 - 0.5 \cdot 200.0 \\
 &= 3468.158 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\
 &= (0.5 \cdot 7021.0 \cdot 36.4 \cdot 822.0 - 2 \cdot 200.0 \cdot 28.0 \cdot 3468.158) / 7840.897 \\
 &= 84.421 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 7021.0 \cdot \cos(0.117) + 81.8 + 84.421); \text{abs}(0.5 \cdot 7021.0 - 84.421)] \\
 &= 3652.579 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.117 - \sin(0.117) \cdot \cos(0.117)) \cdot 36.4 \cdot (7021.0/2)^3 \\
 &= 457929312.0 \text{ cm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (0.45793E+09 + 112.0(3468.158^2 + 200.0^2/12) - 7840.897 \cdot 84.421^2) / 3652.579 \\
 &= 1289078784.000 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 7840.897 / (\pi \cdot 7021.0 \cdot 36.4)) \\
 &= 0.977
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2/3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 0.12891E+10 / (\pi \cdot 7021.0^2 \cdot 36.4)) \\
 &= 0.915
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 7021.0^2 \cdot 36.4 \cdot 1134.7 \\ &= 15990470.0 \text{ kgf-m.} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 7021.0 \cdot 36.4 \cdot 1134.7 \\ &= 9110071.0 \text{ kgf} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(yq)) / (\psi_{i2} \cdot M_{max}) \\ &= 589867 / (0.977 \cdot 9110071) + (\text{abs}(502327) + 589867 \cdot \text{abs}(84.421)) / \\ &\quad (0.915 \cdot 15990470) \\ &= 0.104 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

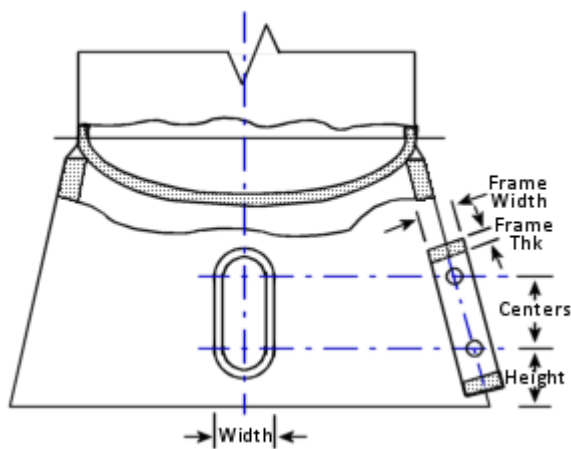
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: SK3

Total vessel (empty) weight	Fg	580783.875 kgf
Head Weight below section 2-2	DeltaFg	260109.078 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	580783.875 kgf
Skirt Mean Diameter	Dz	7021.000 mm
Skirt Analysis Thickness (Corroded)	ez	36.400 mm
Skirt Height	H	1325.000 mm
Additional Axial Force at section 1-1	F1	-9083.000 kgf
Moment at section 1-1	M1	205428.188 kgf-m.
Applied Moment - Bottom of Skirt	Mb	343741.406 kgf-m.
Weight at Bottom of Skirt	W	584954.125 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	7023.000 mm
Head Corroded Thickness	eb	82.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	494718112.000 cm ⁴
Moment of Inertia	Iyy =	494718112.000 cm ⁴
Section Modulus	W =	1409253888.000 mm ³
Section Metal Area	A =	8028.793 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{\frac{\min(Ixx, Iyy)}{\text{Area}}}$
 $= \sqrt{\frac{\min(0.49472E+09, 0.49472E+09)}{8028.793}}$
 $= 2482.298 \text{ mm}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 2482.298 / 1325.0$
 $= 1.873$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 205428.203 kgf-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -205428.203 / .56370E+10 - 584954 / 8028.79346 \\ &= -76.501 \text{ kgf/cm}^2 \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +205428.203 / .56370E+10 - 584954 / 8028.79346 \\ &= -69.213 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F1 - Fg - Ff + 4 \cdot M1 / Dz \\ &= -9083.0 - 580784 - 0.0 + 4 \cdot 205428.203 / 7021.0 \\ &= -472830.6 \text{ kgf} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F1 - Fg - Ff - 4 \cdot M1 / Dz \\ &= -9083.0 - 580784 - 0.0 - 4 \cdot 205428.203 / 7021.0 \\ &= -706903.2 \text{ kgf} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.5E+06 + 260109.078 + 0.0) / (\pi \cdot 7023.0 \cdot 82.0) + 50.76 \cdot 7023.0 / (4 \cdot 82.0) \\ &= 1075.09 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (-0.7E+06 + 260109.078 + 0.0) / (\pi \cdot 7023.0 \cdot 82.0) + 50.76 \cdot 7023.0 / (4 \cdot 82.0) \\ &= 1062.16 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb) \\ &= (0.0 + 260109.078) / (\pi \cdot 7023.0 \cdot 82.0) + 50.76 \cdot 7023.0 / (4 \cdot 82.0) \\ &= 1101.23 \text{ kgf/cm}^2 \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= Fzp / (\pi \cdot Dz \cdot ez) \\ &= -472830.625 / (\pi \cdot 7021.0 \cdot 36.4) \\ &= -58.89 \text{ kgf/cm}^2 \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= Fzq / (\pi \cdot Dz \cdot ez) \\ &= -706903.250 / (\pi \cdot 7021.0 \cdot 36.4) \\ &= -88.05 \text{ kgf/cm}^2 \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$\begin{aligned} &= 1 - 0.5(Db + eb - Dz + ez) / (r + eb) \\ &= 1 - 0.5(7023.0 + 82.0 - 7021.0 + 36.4) / (3470.5 + 82.0) \\ &= 0.9831 \end{aligned}$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.983) = 10.563 \text{ deg}$$

Dimension [a]:

$$\begin{aligned} &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(82.0^2 + 36.4^2 + 2 \cdot 82.0 \cdot 36.4 \cdot 0.983)} \\ &= 58.986 \text{ mm} \end{aligned}$$



Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 58.98 \cdot -472830.6 = -27890.43 \text{ kgf-m.}$$

$$M_q = a \cdot F_{zq} = 58.98 \cdot -706903.2 = -41697.45 \text{ kgf-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{82.0}{36.4} \right)^2$$

$$= 0.341$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.341 \cdot 6 \cdot -27890.43 / (\pi \cdot 7023.0 \cdot 82.0^2)$$

$$= -38.43 \text{ kgf/cm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.341 \cdot 6 \cdot -41697.453 / (\pi \cdot 7023.0 \cdot 82.0^2)$$

$$= -57.46 \text{ kgf/cm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.341 \cdot 6 \cdot -27890.43 / (\pi \cdot 7021.0 \cdot 36.4^2)$$

$$= -195.10 \text{ kgf/cm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.341 \cdot 6 \cdot -41697.453 / (\pi \cdot 7021.0 \cdot 36.4^2)$$

$$= -291.69 \text{ kgf/cm}^2$$

Geometry variables, 16.12-32 to 36:

$$e_b/D_b = 3.228/276.496 = 0.0117$$

$$y = 125 \cdot e_b/D_b = 125 \cdot 82.0/7023.0 = 1.4595$$

$$\alpha = 0$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ kgf/cm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 1075.094 - -38.434 + 0.0$$

$$= 1113.53 \text{ kgf/cm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 1075.094 + -38.434 - 0.0$$

$$= 1036.66 \text{ kgf/cm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 1062.156 - -57.461 + 0.0$$

$$= 1119.62 \text{ kgf/cm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 1062.156 + -57.461 - 0.0$$

$$= 1004.70 \text{ kgf/cm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 1101.229 + -38.434 + 0.0$$

$$= 1062.79 \text{ kgf/cm}^2$$

Stress [σ_{2potot}]:



$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 1101.229 - -38.434 - 0.0 \\
 &= 1139.66 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 1101.229 + -57.461 + 0.0 \\
 &= 1043.77 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 1101.229 - -57.461 - 0.0 \\
 &= 1158.69 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= -58.892 - -195.105 \\
 &= 136.21 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= -58.892 + -195.105 \\
 &= -254.00 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -88.046 - -291.691 \\
 &= 203.64 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -88.046 + -291.691 \\
 &= -379.74 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (1075.094 / 1091.617)^2 \\
 &= 2568.97 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (1062.156 / 1091.617)^2 \\
 &= 2585.86 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (1101.229 / 1091.617)^2 \\
 &= 2534.23 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (1101.229 / 1091.617)^2 \\
 &= 2534.23 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (-58.892 / 1091.617)^2 \\
 &= 3272.73 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 1091.617(3 - (1 / 1.5)) \cdot (-88.046 / 1091.617)^2 \\
 &= 3270.12 \text{ kgf/cm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (kgf/cm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	1113.53	2568.97	Pass	46
Stot1po	1036.66	2568.97	Pass	47
Stot1qi	1119.62	2585.86	Pass	48
Stot1qo	1004.70	2585.86	Pass	49
Stot2pi	1062.79	2534.23	Pass	50
Stot2po	1139.66	2534.23	Pass	51
Stot2qi	1043.77	2534.23	Pass	52
Stot2qo	1158.69	2534.23	Pass	53
Stot3pi	136.21	3272.73	Pass	54
Stot3po	-254.00	3272.73	Pass	55
Stot3qi	203.64	3270.12	Pass	56
Stot3qo	-379.74	3270.12	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Packing is installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	795.864 cm
Center of Gravity of the Packing	725.811 cm
Center of Gravity of the Insulation	725.498 cm
Center of Gravity of the Nozzles	734.061 cm
Center of Gravity of the Added Weights (Operating)	587.759 cm
Center of Gravity of the Added Weights (Empty)	648.212 cm
Center of Gravity of Bare Shell New and Cold	638.206 cm
Center of Gravity of Bare Shell Corroded	639.027 cm
Vessel CG in the Operating Condition	644.262 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	677.334 cm
Vessel CG in the Test Condition	695.107 cm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	7059.0000	mm
Skirt Thickness	STHK	38.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		40.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	1162.42	kgf/cm ²
Basing Yield Stress		2528.76	kgf/cm ²
Inside Diameter of Basing	DI, D4	6898.9995	mm
Outside Diameter of Basing	DOU, D5	7315.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	1407.00	kgf/cm ²
Number of Bolts	RN, nb	32	
Diameter of Bolt Circle	DC, DBC	7187.0000	mm
Bolt Allowable Shear Stress		844.326	kgf/cm ²

Ultimate Comp. Strength of Concrete	FPC	421.8	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	229.4	kgf/cm ²
Modular ratio Steel/Concrete		0.000	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	128.0000	mm
Width of Gussets at Base Plate	BWDT	128.0000	mm
Gusset Plate Elastic Modulus	E	2056000.0	kgf/cm ²
Gusset Plate Yield Stress	SY	2528.8	kgf/cm ²
Height of Chair	h1	300.0000	mm
Distance between Gussets	RG, b6	90.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	38.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm

External Corrosion Allowance	CA	1.6000	mm
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Dead Weight of Vessel	DW	428238.9	kgf
Operating Weight of Vessel	ROW	633238.9	kgf
Earthquake Moment on Basing	EQMOM	748005.8	kgf-m.
Wind Moment on Basing	WIMOM	36836.6	kgf-m.

User Moment at the Base [Earthquake Case]		133676.1	kgf-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		133676.1	kgf-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 881682/7187.0) - 613665)/32$$

$$= -3842.4 \text{ kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 170513/7187.0) - 428239)/32$$

$$= -10416.8 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7187.0) - 633239)/32$$

$$= -19788.7 \text{ kgf}$$

Governing Bolt Load Condition, **** No Tensile Bolt Loads ****

Area Available in 1 Bolt (corroded)	Abss :	8.5561	cm ²
Area Available in all the Bolts	Abss * RN :	273.7959	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.217	0.108	1.271	2.624	0.455	0.777	157964.1	157964.1	8914.0
4	0.054	0.081	0.625	2.997	0.489	0.760	161350.7	161350.7	7970.5
6	0.095	0.088	0.830	2.899	0.481	0.766	160217.3	160217.3	8182.0
8	0.092	0.093	0.814	2.907	0.481	0.765	160301.8	160301.8	8163.9
10	0.092	0.092	0.818	2.905	0.481	0.765	160280.6	160280.6	8168.4
11	0.092	0.092	0.816	2.906	0.481	0.765	160291.2	160291.2	8166.1

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 160291.2 / (1.213 \cdot 7187.0 \cdot 2.906)$$

$$= 1265.754 \text{ kgf/cm}^2, \text{ Should be } \leq 1407.0 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 32 \cdot 8.556 / (\pi \cdot 7187.0)$$

$$= 1.213 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 25.672[(2 \cdot 0.092 \cdot 7187.0 + 208.0) / (2 \cdot 0.092 \cdot 7187.0)]$$

$$= 29.713 \text{ kgf/cm}^2, \text{ Should be } \leq 229.4 \text{ kgf/cm}^2$$

Values for table 10.3, l = 128.0, b = 352.791, l/b = 0.362821

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(429.132, 1989.654) = 1989.654 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 1989.654 / 1669.0)} + 1.6$$

$$= 28.345 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 128.0, b = 90.0, b/l = 0.703125

As b/l (0.703) is less than 1, inverting b/l = 1.422.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 12038.45 / (4 \cdot \pi) [1.3(\ln((2 \cdot 128.0 \cdot \sin(\pi \cdot 64.0/128.0)) / (\pi \cdot 35.0))) + 1] - [(0.7 - 0.08) \cdot 12038.45 / (4 \cdot \pi)]$$

$$= 1416.2883 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(1416.29) / 1685.84) + 1.6}$$

$$= 24.0514 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 12038.45 / (1265.526 \cdot 128.0) + 1.6$$

$$= 11.125 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 7187.0 / 32 = 705.582 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (12038 \cdot 64.0 / (705.58 \cdot 262.0 \cdot 2111))^{2/3} \cdot 3529.51^{1/3} + \text{Ca}$$

$$= 9.087 + 1.6 = 10.687 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 107144.9 / (8.556 \cdot 32)$$

$$= 391.3 \text{ kgf/cm}^2. \text{ Must be less than } 844.3 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	28.3448	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	24.0514	mm
Actual Top Ring Thickness as entered by user	38.0000	mm
Required Gusset thickness, + CA	11.1250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	10.6866	mm
Given Thickness of Skirt	38.0000	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 139.435 \cdot (38.0 - 1.6)$$

$$= 50.75 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 51 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707]$$

$$= 3.549 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 12038.4 / (232.0 + 2 \cdot (224.0 + 38.0))$$

$$= 15.924 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 12038.4 \cdot 64.0 / (232.0 \cdot (262.0) + 0.6667 \cdot (262.0)^2)$$

$$= 7.231 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{15.92^2 + 7.23^2}$$

$$= 17.489 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 17.49 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707]$$



= 1.223 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 12038.4 / (2 \cdot 128.0)$$

$$= 47.025 \text{ kgf/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 12038.4 \cdot 64.0 / (2 \cdot 262.0 \cdot 128.0)$$

$$= 11.487 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{47.03^2 + 11.49^2}$$

$$= 48.408 \text{ kgf/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 48.41 / [(0.4 \cdot 2529) \cdot 2 \cdot 0.707]$$

$$= 3.385 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.3845	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2

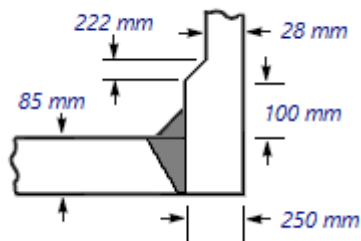
From: 40

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	222.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		250.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3830.0	mm
Centerline Distance to Tangent		4930.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	1.102 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
 $= 81.0658 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.76 \cdot 460.2 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
 $= 22.4496 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 222.0 / 0.5773$
 $= 384.5155 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

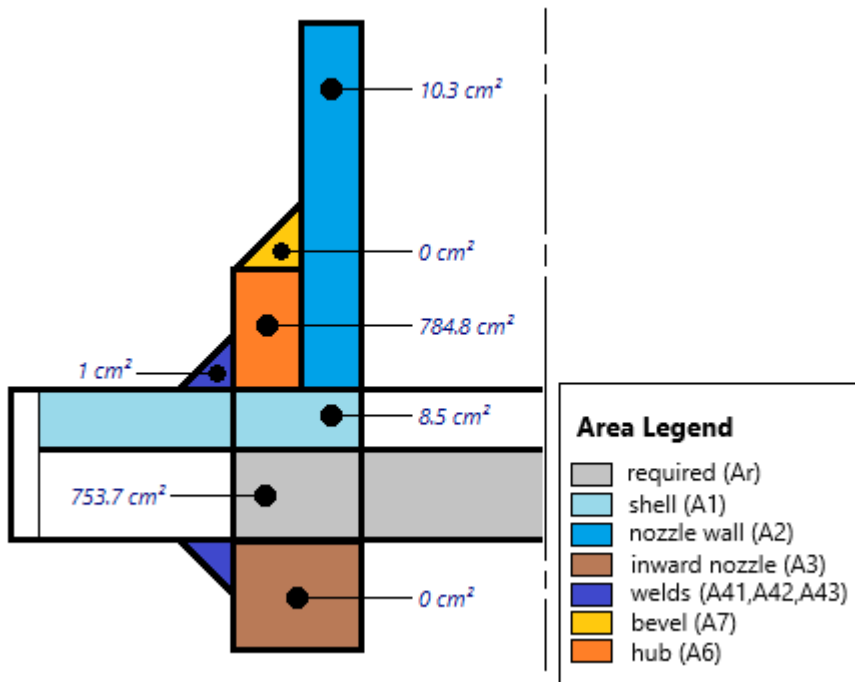
Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	205.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.981, 1.0)$

= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	753.693	NA	NA
Area in Shell	A1	8.511	NA	NA
Area in Nozzle Wall	A2	10.259	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	784.842	NA	NA
TOTAL AREA AVAILABLE	Atot	804.594	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (920.4 \cdot 81.0658 \cdot 1.0 + 2 \cdot 247.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$$

$$= 753.693 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920.4(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 247.0$$

$$(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$$

$$= 8.511 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 205.0)(25.0 - 22.45) \cdot 0.9811$$

$$= 10.259 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

= 0.981 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(205.0, 500.0, 100.0)) \cdot (250.0 - 28.0) \cdot 0.981 \\ &= 784.842 \text{ cm}^2 \text{ Includes Bevel Area of } 349.228 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 25.4496 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 84.0658 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 84.0658 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.331, \max(84.0658, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(25.4496, 11.3312) \\ &= 25.4496 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 602.5,	Allowable	: 1071.0 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2495.0 kgf/cm ²	Passed
Occasional	: 454.8,	Allowable	: 1424.4 kgf/cm ²	Passed
Shear	: 80.4,	Allowable	: 749.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govern. thk, tg = 28.0, tr = 16.97, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.679, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govern. thk, tg = 85.0, tr = 58.564, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.714, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 50.76/105.44 = 0.481$$

Weld Size Calculations, Description: N2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (753.6929 - 8.5114 + 2 \cdot 247.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$
 $= 818394.25 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (795.1013 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092$
 $= 869015.69 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (795.1013 + 0.0 + 0.9811 + (397.4294)) \cdot 1092$
 $= 1302855.88 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (795.1013 + 0.0 + 0.9811 + 0.0 + (397.4294)) \cdot 1092$
 $= 1302855.88 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 1414.4 \cdot 10.0 \cdot 0.49 \cdot 1071$
 $= 116594. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 583.7) \cdot (250.0 - 3.0) \cdot 0.7 \cdot 1071$
 $= 3395654. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 1414.4 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092$
 $= 1471660. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (116594 + 3395654) = 3512248 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (116594 + 0 + 1471660 + 0) = 1588254 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (116594 + 1471660 + 0) = 1588254 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 3512248 kgf, must exceed W = 818394 kgf	or W1 = 869015 kgf
Path 2-2 = 1588253 kgf, must exceed W = 818394 kgf	or W2 = 1302855 kgf
Path 3-3 = 1588253 kgf, must exceed W = 818394 kgf	or W3 = 1302855 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 72.8831 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 657.8830 mm

Input Echo, WRC107/537 Item 1, Description: N2 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	6935.000	mm
Vessel Thickness	Tv	85.000	mm
Design Temperature	T1	475.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1091.62	kgf/cm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	250.000	mm
Nozzle Material		SA-336 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1071.00	kgf/cm ²
Design Internal Pressure	Dp	50.760	kgf/cm ²
Include Pressure Thrust		No	

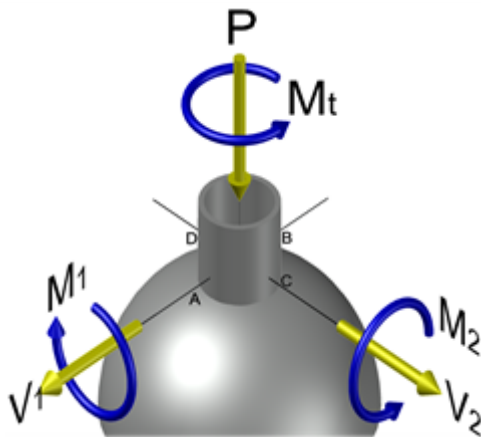
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	8257.0	kgf
Longitudinal Shear (SUS)	(V1) V1	5963.3	kgf
Circumferential Shear (SUS)	(Vc) V2	5963.3	kgf
Circumferential Moment (SUS)	(Mc) M1	16514.4	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	16514.4	kgf-m.
Torsional Moment (SUS)	Mt	18991.5	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 1414.4 + 2 \cdot 1.65 \cdot \sqrt{(3511.5 (85.0 - 3.0))}$$

$$= 3185.190 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	8257.0	kgf
Circumferential Shear	(VC) V2	5963.3	kgf
Longitudinal Shear	(VL) V1	5963.3	kgf
Circumferential Moment	(MC) M1	16514.4	kgf-m.
Longitudinal Moment	(ML) M2	16514.4	kgf-m.
Torsional Moment	MT	18991.5	kgf-m.

Dimensionless Param: U = 1.32 TAU = 5.00 (2.36) RHO = 0.33

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03018	(A,B,C,D)
$M(x) / P$	SP 1	0.03032	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03102	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04589	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03102	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04589	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04473	(A,B,C,D)
$M(y) / P$	SP 1	0.00838	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04220	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01378	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04220	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01378	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7
Rad. Bend.	P	-22.3	22.3	-22.3	22.3	-22.3	22.3	-22.3	22.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-14.2	-14.2	14.2	14.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-126.0	126.0	126.0	-126.0
Rad. Memb.	ML	-14.2	-14.2	14.2	14.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-126.0	126.0	126.0	-126.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-166.3	130.5	114.2	-93.2	-166.3	130.5	114.2	-93.2



Tang. Memb. P	-5.5	-5.5	-5.5	-5.5	-5.5	-5.5	-5.5	-5.5
Tang. Bend. P	-6.2	6.2	-6.2	6.2	-6.2	6.2	-6.2	6.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-19.3	-19.3	19.3	19.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-37.9	37.9	37.9	-37.9
Tang. Memb. ML	-19.3	-19.3	19.3	19.3	0.0	0.0	0.0	0.0
Tang. Bend. ML	-37.9	37.9	37.9	-37.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-68.8	19.2	45.5	-17.9	-68.8	19.2	45.5	-17.9
Shear VC	3.3	3.3	-3.3	-3.3	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
Shear MT	7.4	7.4	7.4	7.4	7.4	7.4	7.4	7.4
Tot. Shear	10.6	10.6	4.1	4.1	4.1	4.1	10.6	10.6
Str. Int.	167.4	131.5	114.4	93.4	166.4	130.6	115.8	94.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Rad.	Pl (SUS)	-17.9	-17.9	10.5	10.5	-17.9	-17.9	10.5	10.5
Rad.	Q (SUS)	-148.4	148.4	103.7	-103.7	-148.4	148.4	103.7	-103.7
Long.	Pm (SUS)	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Long.	Pl (SUS)	-24.8	-24.8	13.8	13.8	-24.8	-24.8	13.8	13.8
Long.	Q (SUS)	-44.0	44.0	31.7	-31.7	-44.0	44.0	31.7	-31.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.3	3.3	-3.3	-3.3	-3.3	-3.3	3.3	3.3
Shear	Q (SUS)	7.4	7.4	7.4	7.4	7.4	7.4	7.4	7.4
Pm (SUS)		1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Pm+Pl (SUS)		1058.0	1058.0	1090.4	1090.4	1058.0	1058.0	1090.4	1090.4
Pm+Pl+Q (Total)		1006.9	1206.0	1189.0	1056.9	1005.9	1205.2	1190.3	1058.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1074.56	1091.62	Passed
Pm+Pl (SUS)	1090.39	1637.43	Passed
Pm+Pl+Q (TOTAL)	1206.02	3274.85	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: UC1

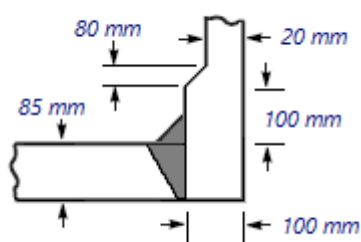
From: 40

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2550.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1850.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	500.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: UC1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 0.787 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 307.8 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 15.0152 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 80.0/0.5773
= 138.5641 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

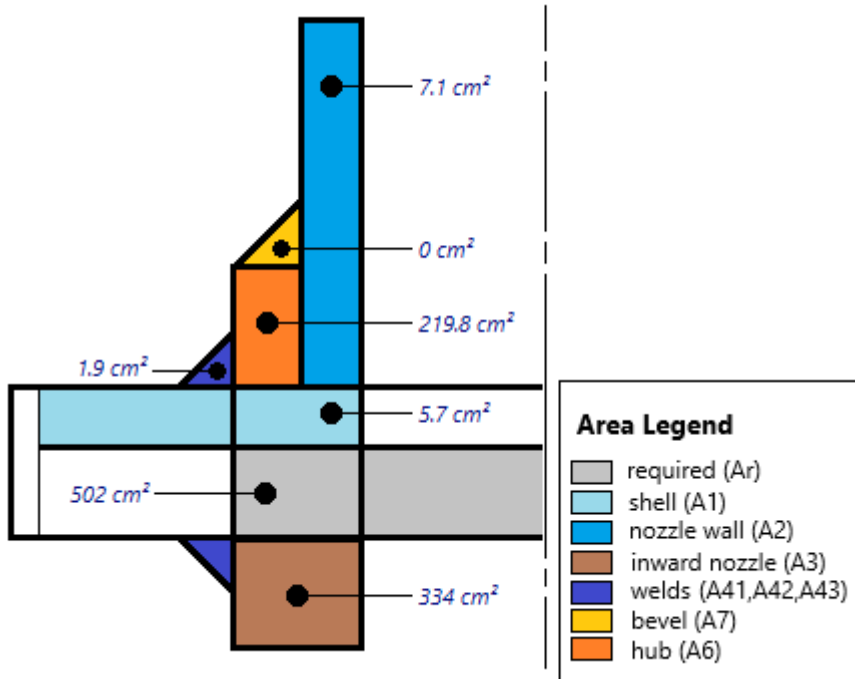
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1231.2001	mm
Parallel to Vessel Wall, opening length	d	615.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	181.0641	mm
Normal to Vessel Wall, Inward		35.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	502.011	NA	NA
Area in Shell	A1	5.717	NA	NA
Area in Nozzle Wall	A2	7.052	NA	NA
Area in Inward Nozzle	A3	333.972	NA	NA
Area in Welds	A41+A42+A43	1.915	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	219.769	NA	NA
TOTAL AREA AVAILABLE	Atot	568.425	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (615.6 \cdot 81.0658 \cdot 1.0 + 2 \cdot 97.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 502.011 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 615.6 (1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 97.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 5.717 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 181.06) (17.0 - 15.02) 0.9811$
 $= 7.052 \text{ cm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 94.0 \cdot (181.0641) \cdot 0.9811$
 $= 333.972 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\ &= 10.0^2 \cdot 0.9811 + (9.7573)^2 \cdot 0.9811 \\ &= 1.915 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(181.1, 1850.0, 100.0)) \cdot (100.0 - 20.0) \cdot 0.9811 \\ &= 219.769 \text{ cm}^2 \text{ Includes Bevel Area of } 62.791 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 18.0152 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 18.0152 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 11.35$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.668$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 85.0$, $t_r = 58.564$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.714$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $50.76 / 105.44 = 0.481$

Weld Size Calculations, Description: UC1

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm
Intermediate Calc. for Inward Weld	T_{minIns}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Inward Weld	$6.0000 = \text{Min per Code}$	$6.8980 = 0.7 \cdot W_i - \text{Can mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) S_v) \\ &= \max(0, (502.0112 - 5.7168 + 2 \cdot 97.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) 1092) \\ &= 543703.62 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wt-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (226.8214 + 1.9152 - 95.2053 \cdot 0.98) \cdot 1092 \\ &= 248672.69 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (226.8214 + 333.9716 + 1.9152 + (156.0756)) \cdot 1092 \\ &= 784635.38 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (226.8214 + 333.9716 + 1.9152 + 0.0 + (156.0756)) \cdot 1092 \\ &= 784635.38 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 809.6 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 66738. \text{ kgf} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 809.6 \cdot 9.7573 \cdot 0.49 \cdot 1071 \\ &= 65119. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 356.3) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 814000. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 809.6 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 842375. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66738 + 814000) = 880738 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (66738 + 0 + 842375 + 65119) = 974232 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (66738 + 842375 + 65119) = 974232 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 880738 kgf, must exceed W = 543703 kgf or W1 = 248672 kgf
Path 2-2 = 974232 kgf, must exceed W = 543703 kgf or W2 = 784635 kgf
Path 3-3 = 974232 kgf, must exceed W = 543703 kgf or W3 = 784635 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.7094 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 2434.9998 mm

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Input, Nozzle Desc: H1

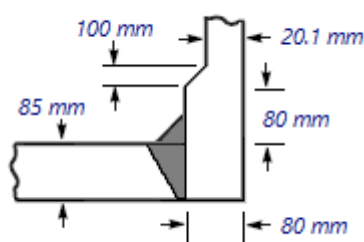
From: 40

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1200.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	317.4500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: H1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 7.040 in.
Actual Thickness Used in Calculation 0.793 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 92.4 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 4.5076 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $59.864 / 0.5773$
= 103.6878 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

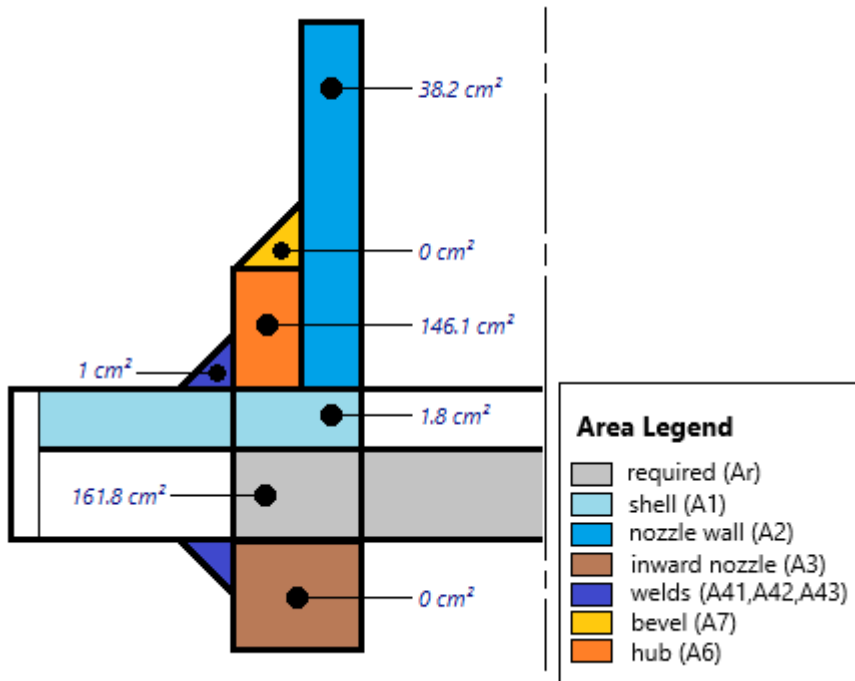
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	394.9527	mm
Parallel to Vessel Wall	Rn+tn+t	197.4764	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	146.5274	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	161.799	NA	NA
Area in Shell	A1	1.825	NA	NA
Area in Nozzle Wall	A2	38.214	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	146.127	NA	NA
TOTAL AREA AVAILABLE	Atot	187.147	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 69.99 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (196.681 \cdot 81.0658 \cdot 1.0 + 2 \cdot 77.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 161.799 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 198.272(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 77.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.825 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 146.53)(17.14 - 4.51)0.9811/\sin(71.8)$
 $= 38.214 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$



= 0.981 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(146.5, 317.5, 80.0)) \cdot (80.0 - 20.1) \cdot 0.981 \\ &= 146.127 \text{ cm}^2 \text{ Includes Bevel Area of } 52.153 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.136, tr = 3.407, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.199, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 80.0, tr = 3.407, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.044, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	45 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.76/105.44 = 0.481

Weld Size Calculations, Description: H1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (161.7988 - 1.8251 + 2 \cdot 77.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092) \\ &= 176170.58 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (184.3403 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\ &= 202299.67 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (184.3403 + 0.0 + 0.9811 + (123.895)) \cdot 1092 \\ &= 337545.38 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (184.3403 + 0.0 + 0.9811 + 0.0 + (123.895)) \cdot 1092 \\ &= 337545.38 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 360.5789 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 29724. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 139.315) \cdot (80.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 252654. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 360.5789 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 375176. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (29724 + 252654) = 282377 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (29724 + 0 + 375176 + 0) = 404900 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (29724 + 375176 + 0) = 404900 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 282377 kgf, must exceed W = 176170 kgf	or W1 = 202299 kgf
Path 2-2 = 404900 kgf, must exceed W = 176170 kgf	or W2 = 337545 kgf
Path 3-3 = 404900 kgf, must exceed W = 176170 kgf	or W3 = 337545 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: H1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.040 in.
Actual Thickness Used in Calculation	0.793 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 92.4 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 4.5076 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

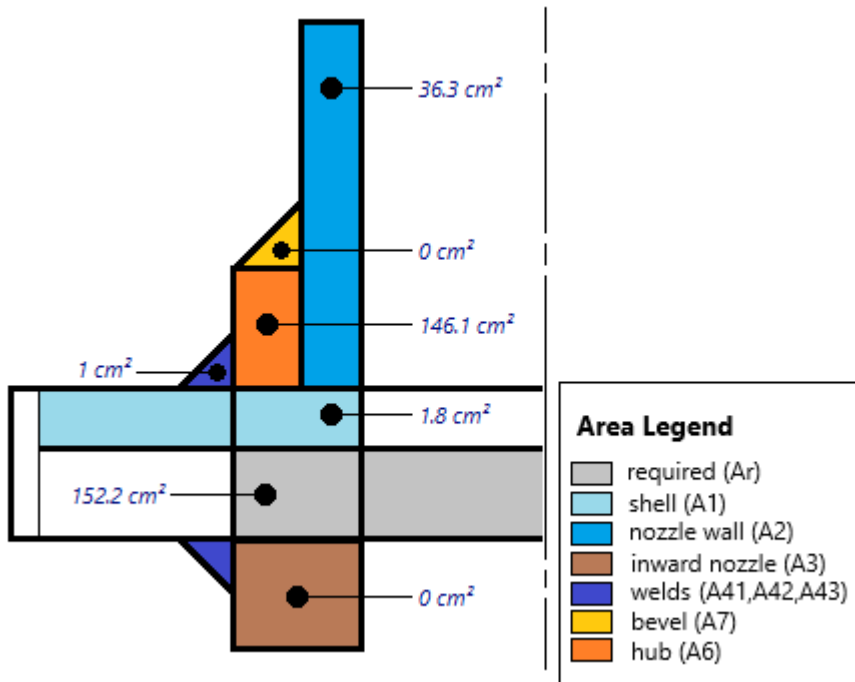
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 59.864 / 0.5773 \end{aligned}$$

= 103.6878 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 383.0750 mm
 Parallel to Vessel Wall Rn+tn+t 191.5375 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 146.5274 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	152.170	NA	NA
Area in Shell	A1	1.825	NA	NA
Area in Nozzle Wall	A2	36.309	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	146.127	NA	NA
TOTAL AREA AVAILABLE	Atot	185.242	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (184.8033 \cdot 81.0658 \cdot 1.0 + 2 \cdot 77.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 152.170 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 198.272(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 77.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.825 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2 \\ &= (2 \cdot 146.53) (17.14 - 4.51) 0.9811 \\ &= 36.309 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811 \\ &= 0.981 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(146.5, 317.5, 80.0)) \cdot (80.0 - 20.1) \cdot 0.9811 \\ &= 146.127 \text{ cm}^2 \text{ Includes Bevel Area of } 52.153 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.5076 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 7.5076 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.1358 mm -> OK

Weld Size Calculations, Description: H1

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v) \\ &= \max(0, (152.17 - 1.8251 + 2 \cdot 77.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) 1092) \\ &= 165659.69 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - can/0.707)^2 \cdot fr_2) \cdot S_v \\ &= (182.4355 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\ &= 200220.38 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (182.4355 + 0.0 + 0.9811 + (123.895)) \cdot 1092 \\ &= 335466.06 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (182.4355 + 0.0 + 0.9811 + 0.0 + (123.895)) \cdot 1092 \\ &= 335466.06 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 338.8033 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 27929. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 130.9016) \cdot (80.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 237396. \text{ kgf} \end{aligned}$$



Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 338.8033 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092$
= 352519. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (27929 + 237396) = 265324 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (27929 + 0 + 352519 + 0) = 380448 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (27929 + 352519 + 0) = 380448 kgf

Summary of Failure Path Calculations:

Path 1-1 = 265324 kgf, must exceed W = 165659 kgf or W1 = 200220 kgf
Path 2-2 = 380448 kgf, must exceed W = 165659 kgf or W2 = 335466 kgf
Path 3-3 = 380448 kgf, must exceed W = 165659 kgf or W3 = 335466 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 67.5990 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 475.5000 mm

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Input, Nozzle Desc: K4

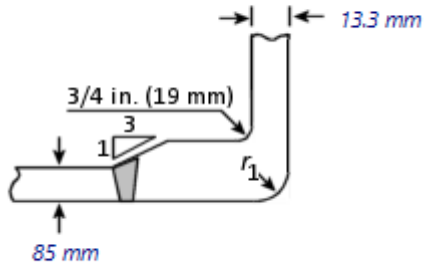
From: 40

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1200.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	267.4500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		f-1	
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



UW-16.1 (f-1)

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.450 in.
Actual Thickness Used in Calculation	0.525 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
 $= 81.0658 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.76 \cdot 34.11 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
 $= 1.6642 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 36.665 / 0.5773$
 $= 63.5057 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

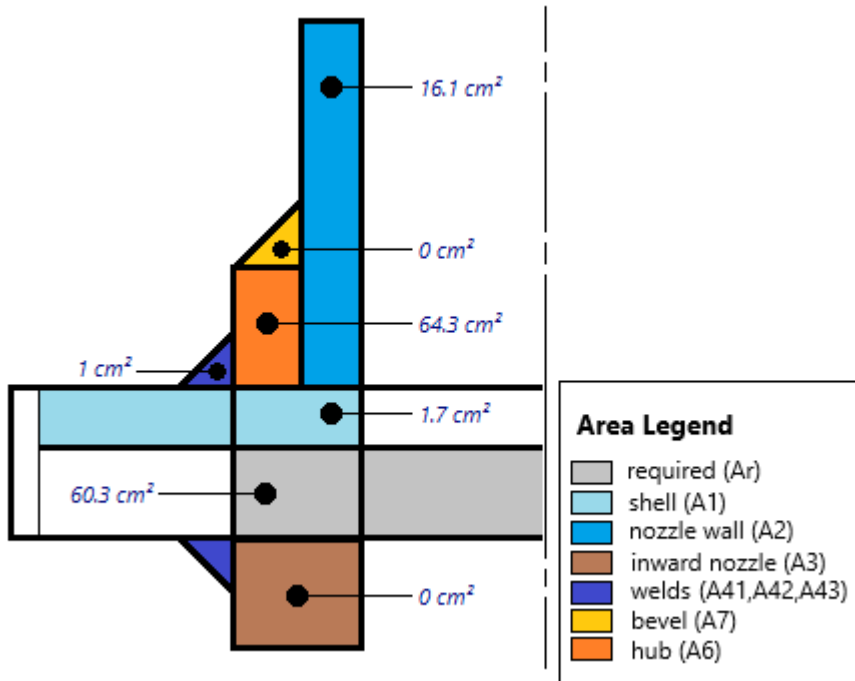
Parallel to Vessel Wall (Diameter Limit)	Dl	257.2742	mm
Parallel to Vessel Wall	Rn+tn+t	128.6371	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	89.3432	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$

= min(0.981, 1.0)
 = 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	60.296	NA	NA
Area in Shell	A1	1.709	NA	NA
Area in Nozzle Wall	A2	16.092	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	64.278	NA	NA
TOTAL AREA AVAILABLE	Atot	83.060	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 70.01 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.6042 \cdot 81.0658 \cdot 1.0 + 2 \cdot 47.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 60.296 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 184.67 (1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 47.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.709 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 89.34) (10.34 - 1.66) 0.9811 / \sin(70.8)$
 $= 16.092 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.



Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\ &= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811 \\ &= 0.981 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(89.3, 267.5, 100.0)) \cdot (50.0 - 13.3) \cdot 0.981 \\ &= 64.278 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 13.335$, $t_r = 1.258$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.122$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 50.0$, $t_r = 1.258$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	36 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $50.76 / 105.44 = 0.481$

Weld Size Calculations, Description: K4

Note:

This nozzle configuration requires a full penetration weld between the nozzle and the shell.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.450 in.
Actual Thickness Used in Calculation	0.525 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, t_r [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
 $= 81.0658$ mm

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.76 \cdot 34.11 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
 $= 1.6642$ mm

Intermediate Hub Nozzle Calculations:

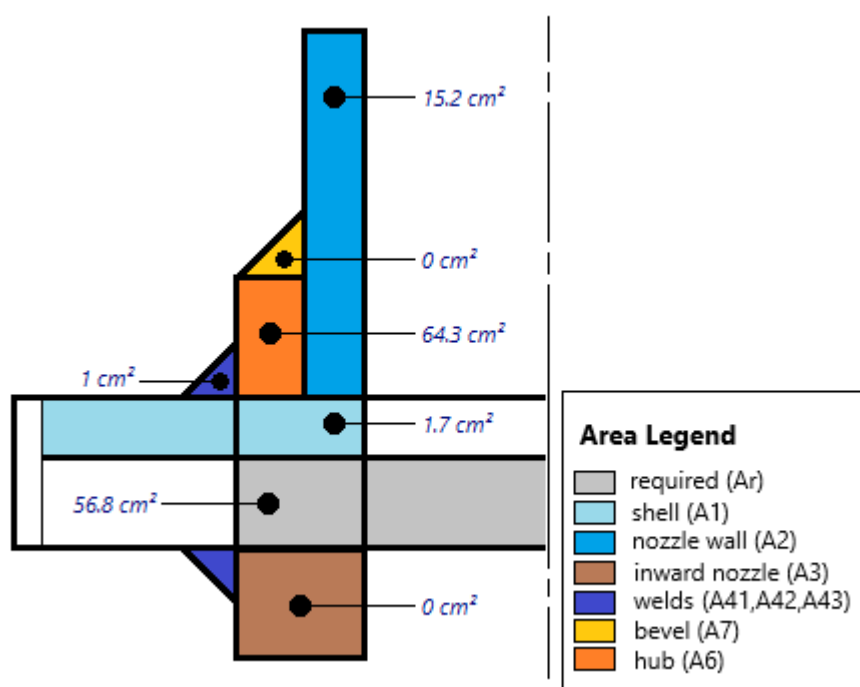
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 36.665/0.5773
 = 63.5057 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	252.9000	mm
Parallel to Vessel Wall	Rn+tn+t	126.4500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	89.3432	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	56.750	NA	NA
Area in Shell	A1	1.709	NA	NA
Area in Nozzle Wall	A2	15.201	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	64.278	NA	NA
TOTAL AREA AVAILABLE	Atot	82.169	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 = (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
 = (68.23•81.0658•1.0+2•47.0•81.0658•1.0•(1-0.98))
 = 56.750 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 184.67(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 47.0 \\ &\quad (1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981) \\ &= 1.709 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 89.34)(10.34 - 1.66)0.9811 \\ &= 15.201 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\ &= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811 \\ &= 0.981 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(89.3, 267.5, 100.0)) \cdot (50.0 - 13.3) \cdot 0.981 \\ &= 64.278 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 4.6642 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= ta \\ &= 4.6642 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 13.3350 mm -> OK

Weld Size Calculations, Description: K4

Note:

This nozzle configuration requires a full penetration weld between the nozzle and the shell.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 31.0801 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.9812 mm

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Input, Nozzle Desc: K2

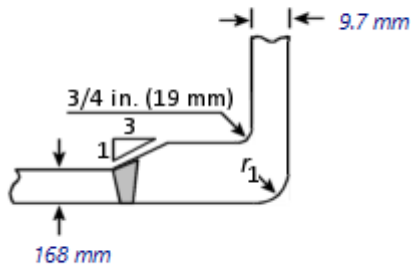
From: 50

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	6860.00	mm
Shell Finished (Minimum) Thickness	t	168.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		95.50	cm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		f-1	
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



UW-16.1 (f-1)

Reinforcement CALCULATION, Description: K2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.612 in.
Actual Thickness Used in Calculation	0.382 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, t_r [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.76 \cdot 3433.0 / (1092 \cdot 1.0 - 0.6 \cdot 50.76)$
 $= 164.2150 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 50.76 \cdot 23.47 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
 $= 1.1450 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 50.31 / 0.5773$
 $= 87.1393 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

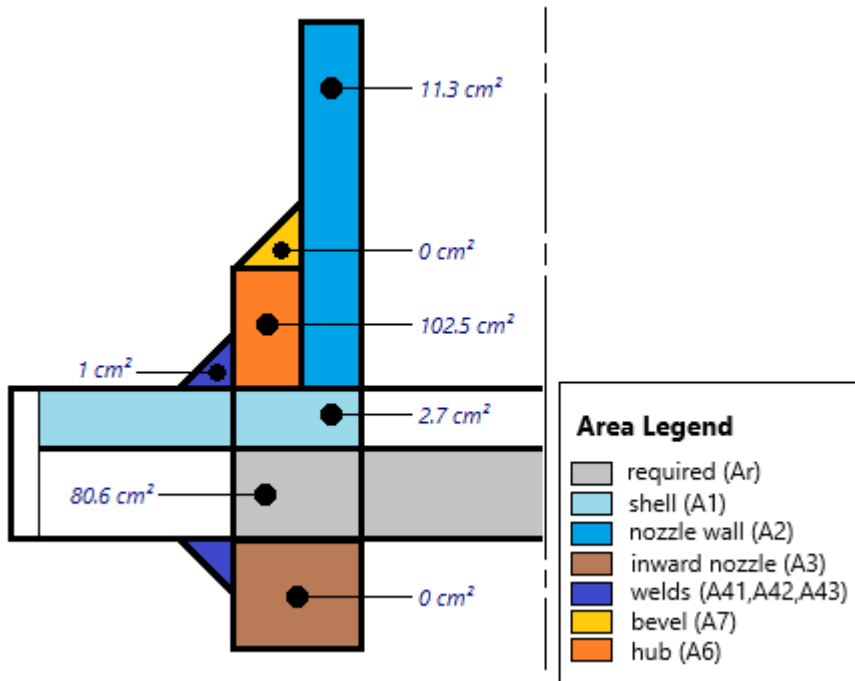
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	390.3250 mm
Parallel to Vessel Wall	$R_n + t_n + t$	195.1625 mm
Normal to Vessel Wall (Thickness Limit), no pad	T_{lnp}	103.8646 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1071.0 / 1091.6)$
 $= 0.981$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.981, 1.0)$
 $= 0.981$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	80.627	NA	NA
Area in Shell	A1	2.679	NA	NA
Area in Nozzle Wall	A2	11.301	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	102.461	NA	NA
TOTAL AREA AVAILABLE	Atot	117.421	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (46.9448 \cdot 164.215 \cdot 1.0 + 2 \cdot 57.0 \cdot 164.215 \cdot 1.0 \cdot (1 - 0.98))$$

$$= 80.627 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 343.38(1.0 \cdot 165.0 - 1.0 \cdot 164.215) - 2 \cdot 57.0$$

$$(1.0 \cdot 165.0 - 1.0 \cdot 164.215) \cdot (1 - 0.981)$$

$$= 2.679 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 103.86)(6.69 - 1.15)0.9811$$

$$= 11.301 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

$$= 0.981 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(103.9, 350.0, 100.0)) \cdot (60.0 - 9.7) \cdot 0.981 \\ &= 102.461 \text{ cm}^2 \text{ Includes Bevel Area of } 3.741 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.1450 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 4.1450 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govern. thk, $t_g = 9.69$, $t_r = 0.866$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.129$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, $t_g = 60.0$, $t_r = 0.866$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.015$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $50.76/105.44 = 0.481$

Weld Size Calculations, Description: K2

Note:

This nozzle configuration requires a full penetration weld between the nozzle and the shell.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9441 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 518.9440 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 195.16 mm

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Input, Nozzle Desc: N1

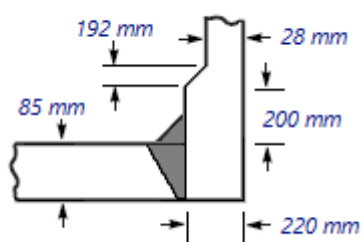
From: 60

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	192.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 1.102 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 460.2 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 22.4496 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 192.0/0.5773
= 332.5539 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

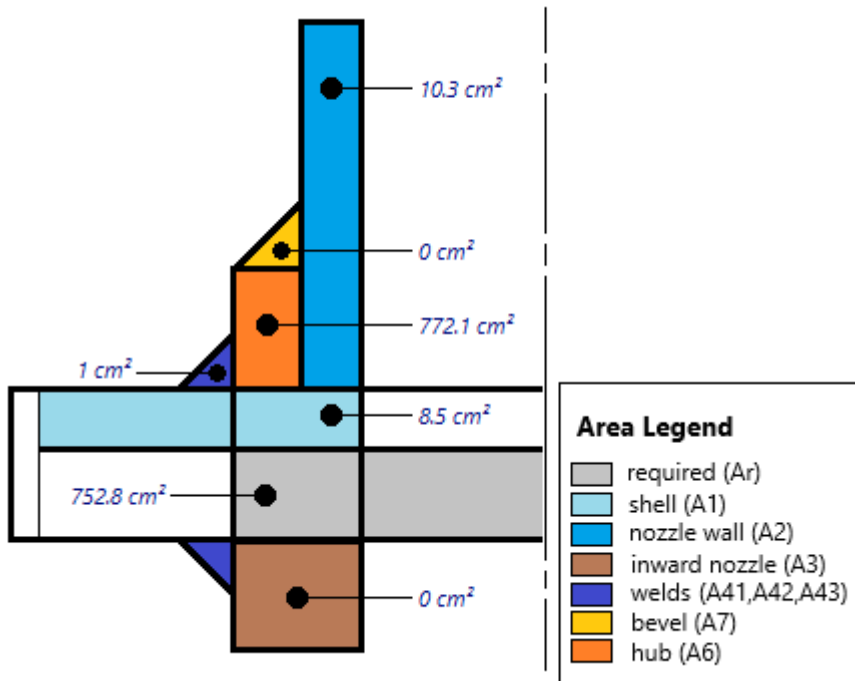
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1840.8000	mm
Parallel to Vessel Wall, opening length	d	920.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	205.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	752.774	NA	NA
Area in Shell	A1	8.522	NA	NA
Area in Nozzle Wall	A2	10.259	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	772.088	NA	NA
TOTAL AREA AVAILABLE	Atot	791.850	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (920.4 \cdot 81.0658 \cdot 1.0 + 2 \cdot 217.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 752.774 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 920.4(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 217.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 8.522 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 205.0)(25.0 - 22.45)0.9811$
 $= 10.259 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$
 $= 0.981 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(205.0, 600.0, 200.0)) \cdot (220.0 - 28.0) \cdot 0.981$
 $= 772.088 \text{ cm}^2$ Includes Bevel Area of 18.593 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 25.4496 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 84.0658 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 84.0658 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(84.0658, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(25.4496, 11.3312)$
 $= 25.4496 \text{ mm}$

Available Nozzle Neck Thickness = 28.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	602.5,	Allowable	:	1071.0 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2495.0 kgf/cm ²	Passed
Occasional	:	454.8,	Allowable	:	1424.4 kgf/cm ²	Passed
Shear	:	80.4,	Allowable	:	749.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 28.0$, $t_r = 16.97$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $t_r \cdot E* / (t_g - c) = 0.679$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 85.0$, $t_r = 58.564$, $c = 3.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio = $t_r \cdot E* / (t_g - c) = 0.714$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $50.76/105.44 = 0.481$

Weld Size Calculations, Description: N1

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
---	------	------------

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (752.7742 - 8.522 + 2 • 217.0 • 0.9811 •
(1.0 • 82.0 - 81.0658))1092)
= 816779.50 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (782.3471 + 0.9811 - 0.0 • 0.98) • 1092
= 855093.06 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (782.3471 + 0.0 + 0.9811 + (349.1586)) • 1092
= 1236240.00 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (782.3471 + 0.0 + 0.9811 + 0.0 + (349.1586)) • 1092
= 1236240.00 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 1354.4 • 10.0 • 0.49 • 1071
= 111648. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 568.7) • (220.0 - 3.0) • 0.7 • 1071
= 2906562. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 1354.4 • (85.0 - 3.0) • 0.74 • 1092
= 1409231. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (111648 + 2906563) = 3018211 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (111648 + 0 + 1409231 + 0) = 1520879 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (111648 + 1409231 + 0) = 1520879 kgf

Summary of Failure Path Calculations:

Path 1-1 = 3018210 kgf, must exceed W = 816779 kgf or W1 = 855093 kgf
Path 2-2 = 1520878 kgf, must exceed W = 816779 kgf or W2 = 1236240 kgf
Path 3-3 = 1520878 kgf, must exceed W = 816779 kgf or W3 = 1236240 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 66.7711 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 751.7709 mm

Input Echo, WRC107/537 Item 1, Description: N1 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	6935.000	mm
Vessel Thickness	Tv	85.000	mm
Design Temperature	T1	475.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	1509.10	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1091.62	kgf/cm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	220.000	mm
Nozzle Material		SA-336 F11	
Nozzle UNS Number		K11572	
Nozzle Cold S.I. Allowable	SNmc	1407.00	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1071.00	kgf/cm ²
Design Internal Pressure	Dp	50.760	kgf/cm ²
Include Pressure Thrust		No	

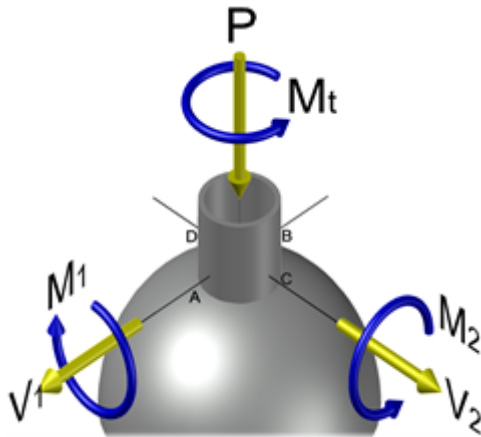
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	8257.0	kgf
Longitudinal Shear (SUS)	(V1) V1	5963.3	kgf
Circumferential Shear (SUS)	(Vc) V2	5963.3	kgf
Circumferential Moment (SUS)	(Mc) M1	16514.4	kgf-m.
Longitudinal Moment (SUS)	(Ml) M2	16514.4	kgf-m.
Torsional Moment (SUS)	Mt	18991.5	kgf-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1354.4 + 2 \cdot 1.65 \cdot \sqrt{(3511.5 (85.0 - 3.0))}$$

$$= 3125.191 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	8257.0	kgf
Circumferential Shear	(VC) V2	5963.3	kgf
Longitudinal Shear	(VL) V1	5963.3	kgf
Circumferential Moment	(MC) M1	16514.4	kgf-m.
Longitudinal Moment	(ML) M2	16514.4	kgf-m.
Torsional Moment	MT	18991.5	kgf-m.

Dimensionless Param: U = 1.26 TAU = 5.00 (2.62) RHO = 0.38

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03179	(A,B,C,D)
$M(x) / P$	SP 1	0.03206	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03342	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.05023	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03342	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.05023	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04788	(A,B,C,D)
$M(y) / P$	SP 1	0.00893	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04600	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01520	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04600	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01520	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9
Rad. Bend.	P	-23.6	23.6	-23.6	23.6	-23.6	23.6	-23.6	23.6
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-15.3	-15.3	15.3	15.3
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-137.9	137.9	137.9	-137.9
Rad. Memb.	ML	-15.3	-15.3	15.3	15.3	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-137.9	137.9	137.9	-137.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-180.8	142.4	125.7	-102.9	-180.8	142.4	125.7	-102.9



Tang. Memb. P	-5.9	-5.9	-5.9	-5.9	-5.9	-5.9	-5.9	-5.9
Tang. Bend. P	-6.6	6.6	-6.6	6.6	-6.6	6.6	-6.6	6.6
Tang. Memb. MC	0.0	0.0	0.0	0.0	-21.1	-21.1	21.1	21.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-41.7	41.7	41.7	-41.7
Tang. Memb. ML	-21.1	-21.1	21.1	21.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-41.7	41.7	41.7	-41.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-75.3	21.4	50.3	-20.0	-75.3	21.4	50.3	-20.0
Shear VC	3.4	3.4	-3.4	-3.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-3.4	-3.4	3.4	3.4
Shear MT	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Tot. Shear	11.5	11.5	4.6	4.6	4.6	4.6	11.5	11.5
Str. Int.	182.0	143.4	126.0	103.2	181.0	142.5	127.4	104.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (kgf/cm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Rad.	Pl (SUS)	-19.2	-19.2	11.4	11.4	-19.2	-19.2	11.4	11.4
Rad.	Q (SUS)	-161.6	161.6	114.3	-114.3	-161.6	161.6	114.3	-114.3
Long.	Pm (SUS)	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Long.	Pl (SUS)	-26.9	-26.9	15.2	15.2	-26.9	-26.9	15.2	15.2
Long.	Q (SUS)	-48.3	48.3	35.2	-35.2	-48.3	48.3	35.2	-35.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.4	3.4	-3.4	-3.4	-3.4	-3.4	3.4	3.4
Shear	Q (SUS)	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Pm (SUS)		1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6	1074.6
Pm+Pl (SUS)		1056.6	1056.6	1091.7	1091.7	1056.6	1056.6	1091.7	1091.7
Pm+Pl+Q (Total)		1000.5	1218.0	1200.6	1054.8	999.5	1217.1	1202.0	1056.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1074.56	1091.62	Passed
Pm+Pl (SUS)	1091.75	1637.43	Passed
Pm+Pl+Q (TOTAL)	1218.00	3274.85	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M1

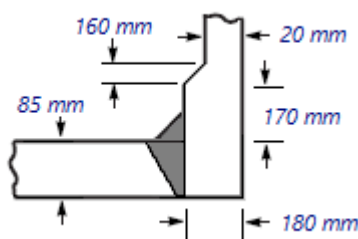
From: 60

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1800.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	170.0000	mm
Height of Beveled Transition	L'	160.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	750.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: M1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 307.8 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 15.0152 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 160.0/0.5773
= 277.1283 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

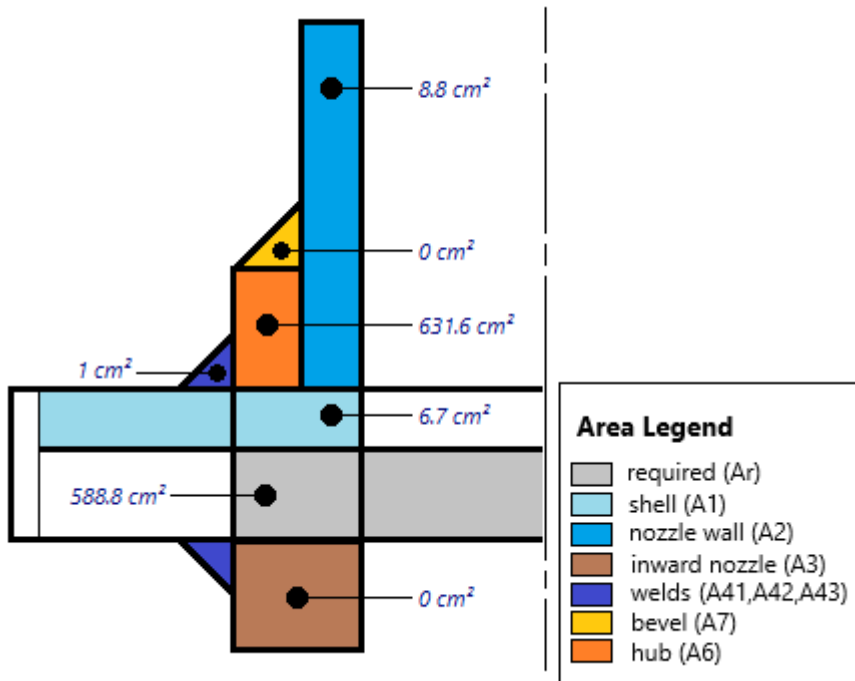
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1439.2145	mm
Parallel to Vessel Wall, opening length	d	719.6072	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	205.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	588.775	NA	NA
Area in Shell	A1	6.660	NA	NA
Area in Nozzle Wall	A2	8.798	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	631.592	NA	NA
TOTAL AREA AVAILABLE	Atot	648.031	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 58.81 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (719.6072 \cdot 81.0658 \cdot 1.0 + 2 \cdot 177.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$$

$$= 588.775 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 719.607 (1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 177.0$$

$$(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$$

$$= 6.660 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 205.0) (17.0 - 15.02) 0.9811 / \sin(65.2)$$

$$= 8.798 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

= 0.981 cm²

Area Available in the Hub Section [A6]:
= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2
= (2 • min(205.0,750.0,170.0)) • (180.0 - 20.0) • 0.981)
= 631.592 cm² Includes Bevel Area of 97.866 cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20.0, tr = 11.35, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.668, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 85.0, tr = 58.564, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.714, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.76/105.44 = 0.481

Weld Size Calculations, Description: M1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (588.7751 - 6.6603 + 2 • 177.0 • 0.9811 •
(1.0 • 82.0 - 81.0658))1092)
= 638987.50 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (640.39 + 0.9811 - 0.0 • 0.98) • 1092
= 700130.44 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (640.39 + 0.0 + 0.9811 + (284.7976)) • 1092
= 1011019.81 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (640.39 + 0.0 + 0.9811 + 0.0 + (284.7976)) • 1092
= 1011019.81 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 1133.4165 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 93432. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 463.2559) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 1931217. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 1133.4165 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 1179301. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (93432 + 1931217) = 2024649 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (93432 + 0 + 1179301 + 0) = 1272733 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (93432 + 1179301 + 0) = 1272733 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 2024648 kgf, must exceed W = 638987 kgf	or W1 = 700130 kgf
Path 2-2 = 1272732 kgf, must exceed W = 638987 kgf	or W2 = 1011019 kgf
Path 3-3 = 1272732 kgf, must exceed W = 638987 kgf	or W3 = 1011019 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: M1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 307.8 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 15.0152 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:



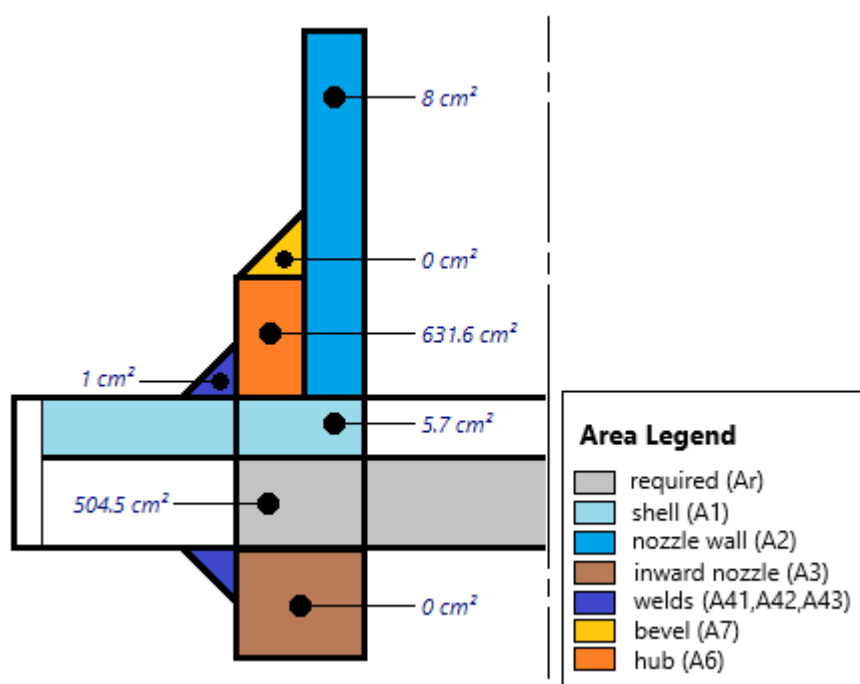
Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 160.0/0.5773
 = 277.1283 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1231.2001 mm
 Parallel to Vessel Wall, opening length d 615.6000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 205.0000 mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	504.461	NA	NA
Area in Shell	A1	5.689	NA	NA
Area in Nozzle Wall	A2	7.984	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	631.592	NA	NA
TOTAL AREA AVAILABLE	Atot	646.246	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 = (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
 = (615.6•81.0658•1.0+2•177.0•81.0658•1.0•(1-0.98))
 = 504.461 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 615.6(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 177.0$$

$$(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$$

$$= 5.689 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 205.0)(17.0 - 15.02)0.9811$$

$$= 7.984 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

$$= 0.981 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(205.0, 750.0, 170.0)) \cdot (180.0 - 20.0) \cdot 0.9811$$

$$= 631.592 \text{ cm}^2 \text{ Includes Bevel Area of } 97.866 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 18.0152 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 18.0152 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Weld Size Calculations, Description: M1

Intermediate Calc. for nozzle/shell welds $Tmin = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot Wo \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (504.4609 - 5.6886 + 2 \cdot 177.0 \cdot 0.9811 \cdot$$

$$(1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$$

$$= 548009.44 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$$

$$= (639.5762 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092$$

$$= 699242.19 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (639.5762 + 0.0 + 0.9811 + (284.7976)) \cdot 1092$$

$$= 1010131.56 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (639.5762 + 0.0 + 0.9811 + 0.0 + (284.7976)) \cdot 1092$$

$$= 1010131.56 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 969.6 \cdot 10.0 \cdot 0.49 \cdot 1071$$

$$= 79928. \text{ kgf}$$



Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 396.3) \cdot (180.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 1652092. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 969.6 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 1008853. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (79928 + 1652092) = 1732019 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (79928 + 0 + 1008853 + 0) = 1088780 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (79928 + 1008853 + 0) = 1088780 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1732019 kgf, must exceed W = 548009 kgf	or W1 = 699242 kgf
Path 2-2 = 1088780 kgf, must exceed W = 548009 kgf	or W2 = 1010131 kgf
Path 3-3 = 1088780 kgf, must exceed W = 548009 kgf	or W3 = 1010131 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 355.4035 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1204.4171 mm

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From: 60

Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.612 in.
Actual Thickness Used in Calculation 0.382 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 23.47 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 1.1450 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $30.31 / 0.5773$
= 52.4983 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

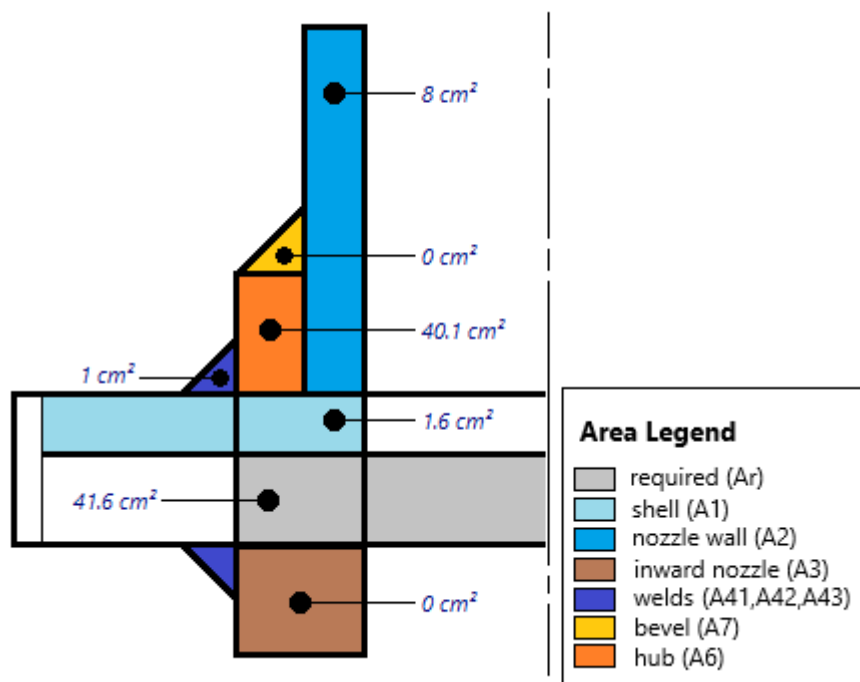
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	227.3335	mm
Parallel to Vessel Wall	Rn+tn+t	113.6667	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	69.2236	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	41.628	NA	NA
Area in Shell	A1	1.644	NA	NA
Area in Nozzle Wall	A2	7.982	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	40.072	NA	NA
TOTAL AREA AVAILABLE	Atot	50.679	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 70.01 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (49.9533 \cdot 81.0658 \cdot 1.0 + 2 \cdot 37.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 41.628 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 177.38(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 37.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.644 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 69.22)(6.69 - 1.15)0.9811/\sin(70.7)$
 $= 7.982 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$

= 0.981 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(69.2, 850.0, 50.0)) \cdot (40.0 - 9.7) \cdot 0.981$$

$$= 40.072 \text{ cm}^2 \text{ Includes Bevel Area of } 10.334 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.69, tr = 0.866, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.129, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40.0, tr = 0.866, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.023, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 31 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 50.76/105.44 = 0.481

Weld Size Calculations, Description: K1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (41.628 - 1.6441 + 2 \cdot 37.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$$

$$= 44387.45 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (48.0543 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092$$

$$= 53527.77 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (48.0543 + 0.0 + 0.9811 + (59.534)) \cdot 1092$$

$$= 118515.95 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (48.0543 + 0.0 + 0.9811 + 0.0 + (59.534)) \cdot 1092$$

$$= 118515.95 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 128.6955 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 10609. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 44.6622) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 38921. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 128.6955 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 133906. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10609 + 38921) = 49529 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10609 + 0 + 133906 + 0) = 144514 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (10609 + 133906 + 0) = 144514 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 49529	kgf, must exceed W = 44387	kgf	or W1 = 53527	kgf
Path 2-2 = 144514	kgf, must exceed W = 44387	kgf	or W2 = 118515	kgf
Path 3-3 = 144514	kgf, must exceed W = 44387	kgf	or W3 = 118515	kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.612	in.
Actual Thickness Used in Calculation	0.382	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 23.47 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 1.1450 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

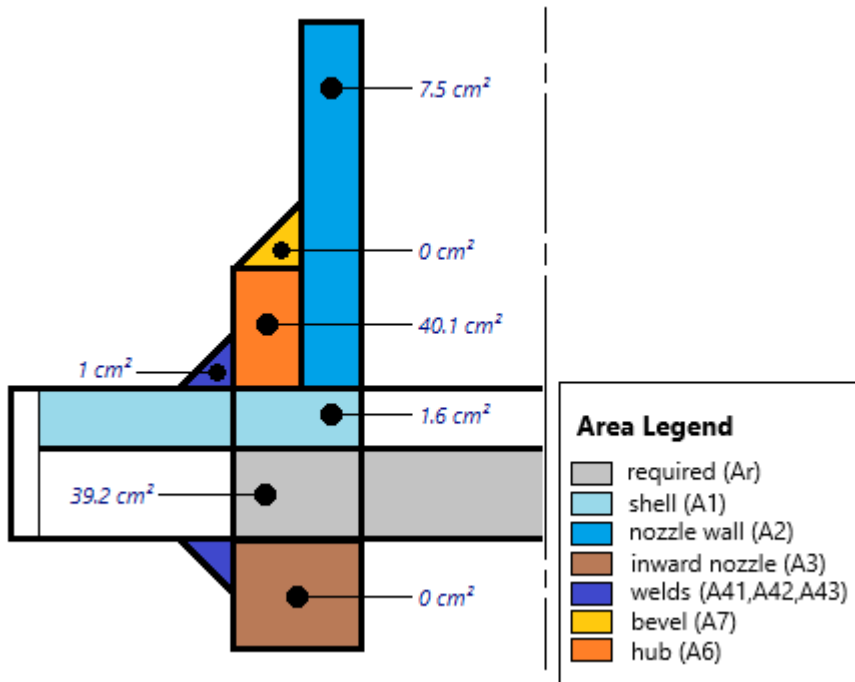
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 30.31 / 0.5773 \end{aligned}$$

= 52.4983 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	224.3250	mm
Parallel to Vessel Wall	Rn+tn+t	112.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	69.2236	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.189	NA	NA
Area in Shell	A1	1.644	NA	NA
Area in Nozzle Wall	A2	7.532	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	40.072	NA	NA
TOTAL AREA AVAILABLE	Atot	50.229	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (46.9448 \cdot 81.0658 \cdot 1.0 + 2 \cdot 37.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 39.189 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 177.38(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 37.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.644 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$\begin{aligned}
 &= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr2 \\
 &= (2 \cdot 69.22) (6.69 - 1.15) \cdot 0.9811 \\
 &= 7.532 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811 \\
 &= 0.981 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr2 \\
 &= (2 \cdot \min(69.2, 850.0, 50.0)) \cdot (40.0 - 9.7) \cdot 0.9811 \\
 &= 40.072 \text{ cm}^2 \text{ Includes Bevel Area of } 10.334 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.1450 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= t_a \\
 &= 4.1450 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Weld Size Calculations, Description: K1

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v) \\
 &= \max(0, (39.1891 - 1.6441 + 2 \cdot 37.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092) \\
 &= 41725.19 \text{ kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - can/0.707)^2 \cdot fr2) \cdot S_v \\
 &= (47.6038 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\
 &= 53036.02 \text{ kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v \\
 &= (47.6038 + 0.0 + 0.9811 + (59.534)) \cdot 1092 \\
 &= 118024.20 \text{ kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\
 &= (47.6038 + 0.0 + 0.9811 + 0.0 + (59.534)) \cdot 1092 \\
 &= 118024.20 \text{ kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 120.9448 \cdot 10.0 \cdot 0.49 \cdot 1071 \\
 &= 9970. \text{ kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 41.9724) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 1071 \\
 &= 36577. \text{ kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 120.9448 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092$
= 125841. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (9970 + 36577) = 46546 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (9970 + 0 + 125841 + 0) = 135811 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (9970 + 125841 + 0) = 135811 kgf

Summary of Failure Path Calculations:

Path 1-1 = 46546 kgf, must exceed W = 41725 kgf	or W1 = 53036 kgf
Path 2-2 = 135810 kgf, must exceed W = 41725 kgf	or W2 = 118024 kgf
Path 3-3 = 135810 kgf, must exceed W = 41725 kgf	or W3 = 118024 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 22.9492 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 963.4003 mm

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Input, Nozzle Desc: K3

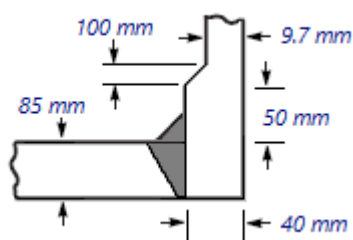
From: 60

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1075.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	826.8481	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K3



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.612 in.
Actual Thickness Used in Calculation 0.382 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 23.47 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 1.1450 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $30.31 / 0.5773$
= 52.4983 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

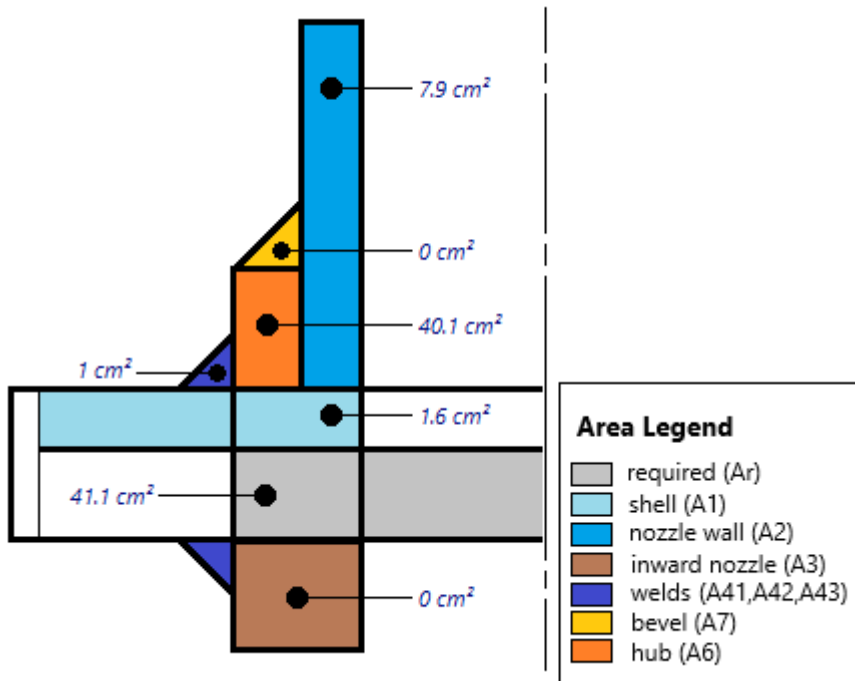
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	226.6939	mm
Parallel to Vessel Wall	Rn+tn+t	113.3470	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	69.2236	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, $1071.0 / 1091.6$)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	41.110	NA	NA
Area in Shell	A1	1.644	NA	NA
Area in Nozzle Wall	A2	7.885	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	40.072	NA	NA
TOTAL AREA AVAILABLE	Atot	50.582	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.17 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (49.3137 \cdot 81.0658 \cdot 1.0 + 2 \cdot 37.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 41.110 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 177.38 (1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 37.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.644 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 69.22) (6.69 - 1.15) 0.9811 / \sin(72.8)$
 $= 7.885 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$



= 0.981 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(69.2, 826.8, 50.0)) \cdot (40.0 - 9.7) \cdot 0.981 \\ &= 40.072 \text{ cm}^2 \text{ Includes Bevel Area of } 10.334 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.69, tr = 0.866, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.129, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40.0, tr = 0.866, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.023, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 31 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 50.76/105.44 = 0.481

Weld Size Calculations, Description: K3

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (41.1095 - 1.6441 + 2 \cdot 37.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092) \\ &= 43821.50 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i \cdot Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (47.9571 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\ &= 53421.73 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (47.9571 + 0.0 + 0.9811 + (59.534)) \cdot 1092 \\ &= 118409.93 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (47.9571 + 0.0 + 0.9811 + 0.0 + (59.534)) \cdot 1092 \\ &= 118409.93 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 127.0479 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 10473. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 44.0904) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 38422. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 127.0479 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 132191. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (10473 + 38422) = 48895 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10473 + 0 + 132191 + 0) = 142664 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (10473 + 132191 + 0) = 142664 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 48895	kgf, must exceed W = 43821 kgf	or W1 = 53421 kgf
Path 2-2 = 142664	kgf, must exceed W = 43821 kgf	or W2 = 118409 kgf
Path 3-3 = 142664	kgf, must exceed W = 43821 kgf	or W3 = 118409 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.612 in.
Actual Thickness Used in Calculation	0.382 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 23.47 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 1.1450 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

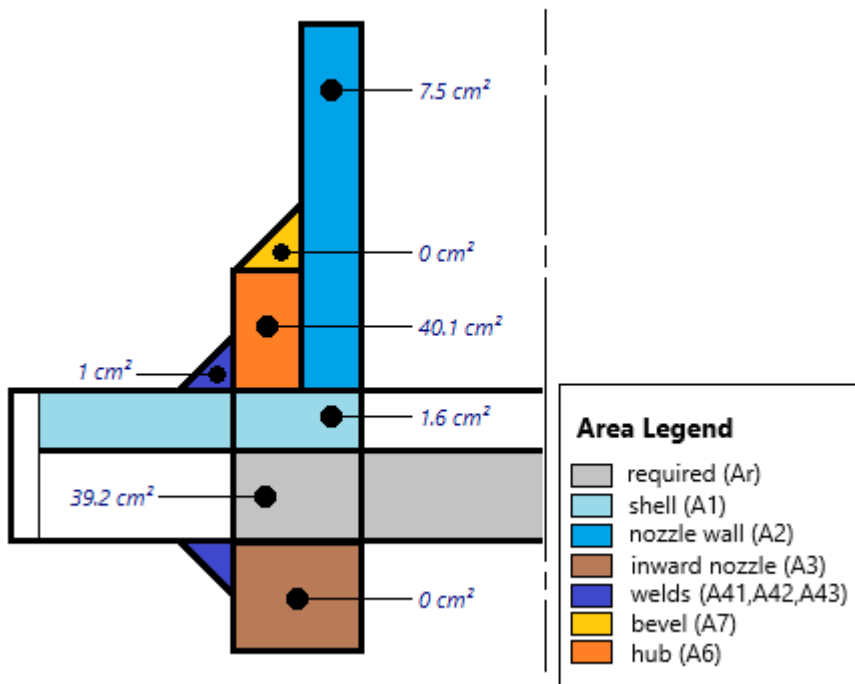
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 30.31 / 0.5773 \end{aligned}$$

= 52.4983 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	224.3250	mm
Parallel to Vessel Wall	Rn+tn+t	112.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	69.2236	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.189	NA	NA
Area in Shell	A1	1.644	NA	NA
Area in Nozzle Wall	A2	7.532	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	40.072	NA	NA
TOTAL AREA AVAILABLE	Atot	50.229	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (46.9448 \cdot 81.0658 \cdot 1.0 + 2 \cdot 37.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 39.189 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 177.38(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 37.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.644 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) fr_2$$

$$= (2 \cdot 69.22) (6.69 - 1.15) 0.9811$$

$$= 7.532 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

$$= 0.981 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr_2$$

$$= (2 \cdot \min(69.2, 826.8, 50.0)) \cdot (40.0 - 9.7) \cdot 0.9811$$

$$= 40.072 \text{ cm}^2 \text{ Includes Bevel Area of } 10.334 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.1450 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 84.0658 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 84.0658 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.4200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.42, \max(84.0658, 4.5)]$$

$$= 6.4200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.145, 6.42)$$

$$= 6.4200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6901 mm -> OK

Weld Size Calculations, Description: K3

Intermediate Calc. for nozzle/shell Welds	$T_{min} = 19.0000 \text{ mm}$
---	--------------------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (39.1891 - 1.6441 + 2 \cdot 37.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$$

$$= 41725.19 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (47.6038 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092$$

$$= 53036.02 \text{ kgf}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (47.6038 + 0.0 + 0.9811 + (59.534)) \cdot 1092$$

$$= 118024.20 \text{ kgf}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (47.6038 + 0.0 + 0.9811 + 0.0 + (59.534)) \cdot 1092$$

$$= 118024.20 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 120.9448 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 9970. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 41.9724) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 36577. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 120.9448 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 125841. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9970 + 36577) = 46546 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (9970 + 0 + 125841 + 0) = 135811 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (9970 + 125841 + 0) = 135811 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 46546 kgf, must exceed W = 41725 kgf	or W1 = 53036 kgf
Path 2-2 = 135810 kgf, must exceed W = 41725 kgf	or W2 = 118024 kgf
Path 3-3 = 135810 kgf, must exceed W = 41725 kgf	or W3 = 118024 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 20.3367 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 936.4763 mm

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Input, Nozzle Desc: K5

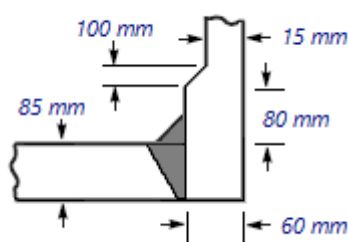
From: 60

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1600.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	100.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1026.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.320 in.
Actual Thickness Used in Calculation 0.590 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 45.17 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 2.2035 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 45.02/0.5773
= 77.9776 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

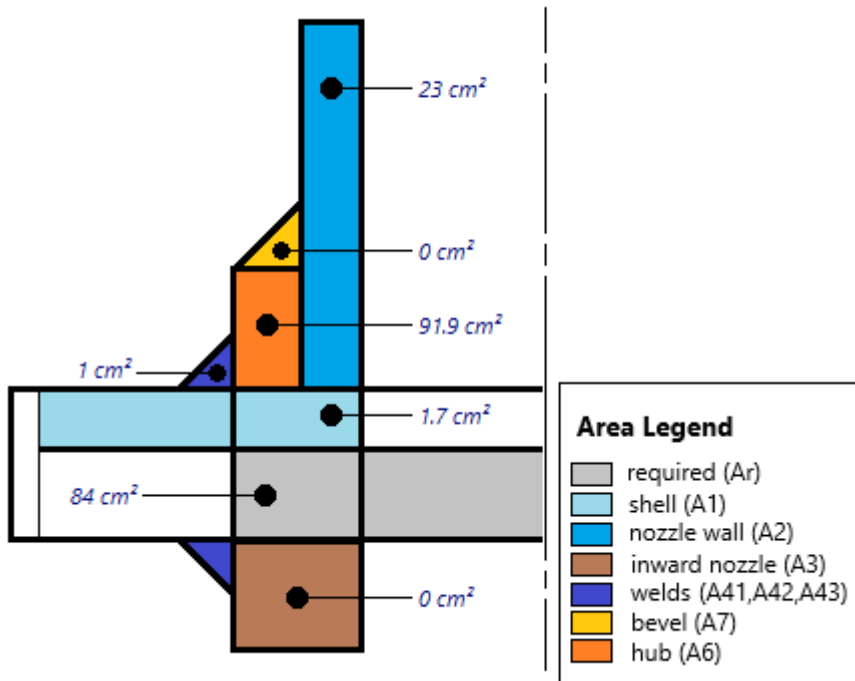
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	289.4571	mm
Parallel to Vessel Wall	Rn+tn+t	144.7286	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	107.9267	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	84.025	NA	NA
Area in Shell	A1	1.736	NA	NA
Area in Nozzle Wall	A2	23.026	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	91.898	NA	NA
TOTAL AREA AVAILABLE	Atot	117.641	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 62.88 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (101.4978 \cdot 81.0658 \cdot 1.0 + 2 \cdot 57.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 84.025 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 187.959(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 57.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.736 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 107.93)(11.98 - 2.2)0.9811/\sin(64.0)$
 $= 23.026 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$

= 0.981 cm²

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(107.9, 1026.8, 80.0)) \cdot (60.0 - 15.0) \cdot 0.981$$

$$= 91.898 \text{ cm}^2 \text{ Includes Bevel Area of } 21.226 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 14.98, tr = 1.666, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.139$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 60.0, tr = 1.666, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.029$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $50.76 / 105.44 = 0.481$

Weld Size Calculations, Description: K5

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (84.0254 - 1.7358 + 2 \cdot 57.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$$

$$= 90969.19 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (114.924 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092$$

$$= 126523.75 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (114.924 + 0.0 + 0.9811 + (91.7145)) \cdot 1092$$

$$= 226640.70 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (114.924 + 0.0 + 0.9811 + 0.0 + (91.7145)) \cdot 1092$$

$$= 226640.70 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 229.5769 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 18925. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 82.7687) \cdot (60.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 111116. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 229.5769 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 238871. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (18925 + 111116) = 130041 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (18925 + 0 + 238871 + 0) = 257796 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (18925 + 238871 + 0) = 257796 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 130041 kgf, must exceed W = 90969 kgf	or W1 = 126523 kgf
Path 2-2 = 257795 kgf, must exceed W = 90969 kgf	or W2 = 226640 kgf
Path 3-3 = 257795 kgf, must exceed W = 90969 kgf	or W3 = 226640 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.320 in.
Actual Thickness Used in Calculation	0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 45.17 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 2.2035 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

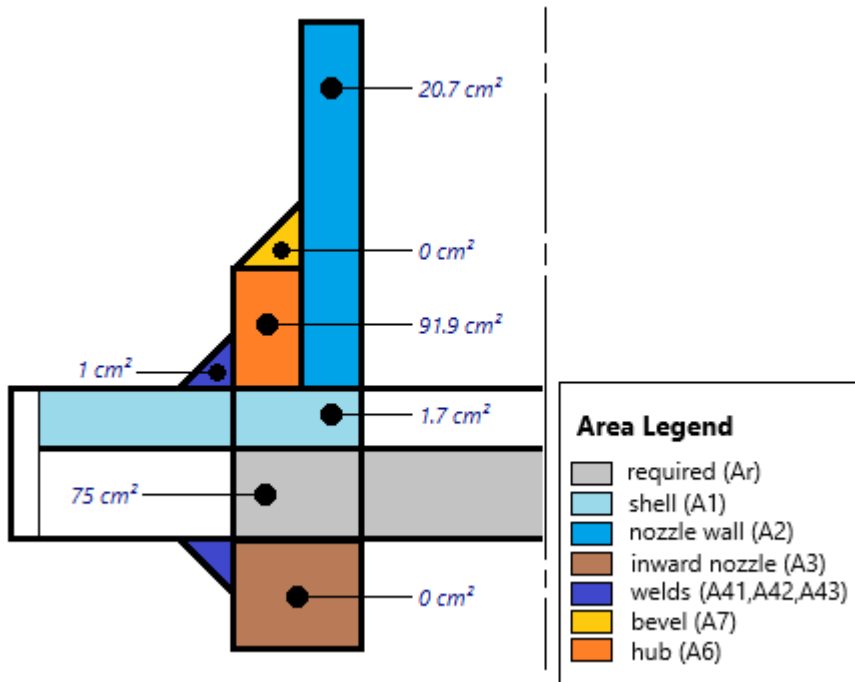
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 45.02/0.5773 \end{aligned}$$

= 77.9776 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	278.3000	mm
Parallel to Vessel Wall	Rn+tn+t	139.1500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	107.9267	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	74.981	NA	NA
Area in Shell	A1	1.736	NA	NA
Area in Nozzle Wall	A2	20.704	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	91.898	NA	NA
TOTAL AREA AVAILABLE	Atot	115.318	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (90.3407 \cdot 81.0658 \cdot 1.0 + 2 \cdot 57.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 74.981 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 187.959(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 57.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.736 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr2$$

$$= (2 \cdot 107.93) (11.98 - 2.2) \cdot 0.9811$$

$$= 20.704 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$$

$$= 0.981 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr2$$

$$= (2 \cdot \min(107.9, 1026.8, 80.0)) \cdot (60.0 - 15.0) \cdot 0.9811$$

$$= 91.898 \text{ cm}^2 \text{ Includes Bevel Area of } 21.226 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.2035 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 5.2035 \text{ mm}$$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Weld Size Calculations, Description: K5

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (74.9808 - 1.7358 + 2 \cdot 57.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092)$$

$$= 81095.93 \text{ kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - can/0.707)^2 \cdot fr2) \cdot S_v$$

$$= (112.6014 + 0.9811 - 0.0 \cdot 0.9811) \cdot 1092$$

$$= 123988.42 \text{ kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (112.6014 + 0.0 + 0.9811 + (91.7145)) \cdot 1092$$

$$= 224105.38 \text{ kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (112.6014 + 0.0 + 0.9811 + 0.0 + (91.7145)) \cdot 1092$$

$$= 224105.38 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 204.3407 \cdot 10.0 \cdot 0.49 \cdot 1071$$

$$= 16845. \text{ kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 73.6703) \cdot (60.0 - 3.0) \cdot 0.7 \cdot 1071$$

$$= 98902. \text{ kgf}$$



Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 204.3407 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092$
= 212613. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (16845 + 98902) = 115746 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (16845 + 0 + 212613 + 0) = 229458 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (16845 + 212613 + 0) = 229458 kgf

Summary of Failure Path Calculations:

Path 1-1 = 115746 kgf, must exceed W = 81095 kgf	or W1 = 123988 kgf
Path 2-2 = 229457 kgf, must exceed W = 81095 kgf	or W2 = 224105 kgf
Path 3-3 = 229457 kgf, must exceed W = 81095 kgf	or W3 = 224105 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 55.3338 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 1177.6849 mm

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Input, Nozzle Desc: K6

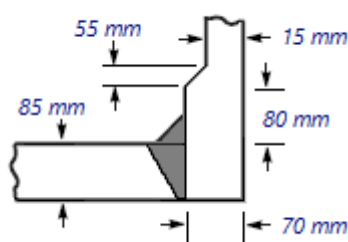
From: 60

Pressure for Reinforcement Calculations	P	50.760	kgf/cm ²
Temperature for Internal Pressure	Temp	475	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	1091.62	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1509.10	kgf/cm ²
Inside Diameter of Hemispherical Head	D	6935.00	mm
Head Finished (Minimum) Thickness	t	85.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	3000.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1071.00	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.00	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	55.0203	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	2200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		84.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.



Reinforcement CALCULATION, Description: K6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.320 in.
Actual Thickness Used in Calculation 0.590 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76)$
= 81.0658 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $50.76 \cdot 45.17 / (1071 \cdot 1.0 - 0.6 \cdot 50.76)$
= 2.2035 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 55.02/0.5773
= 95.2981 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

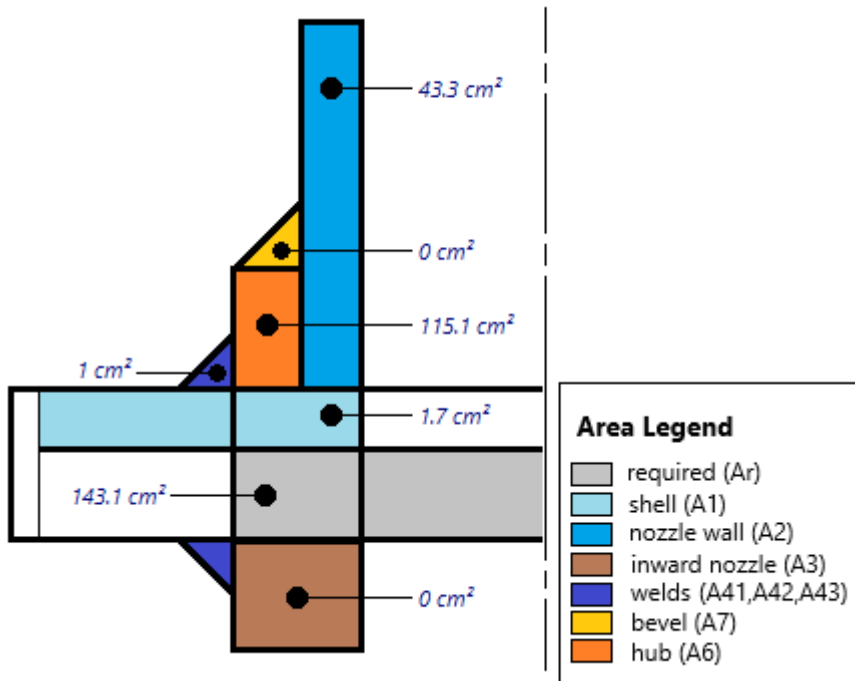
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	362.0094	mm
Parallel to Vessel Wall	Rn+tn+t	181.0047	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.2472	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 1071.0/1091.6)
= 0.981

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.981, 1.0)
= 0.981



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	143.147	NA	NA
Area in Shell	A1	1.732	NA	NA
Area in Nozzle Wall	A2	43.280	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	115.133	NA	NA
TOTAL AREA AVAILABLE	Atot	161.127	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 31.27 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (174.0502 \cdot 81.0658 \cdot 1.0 + 2 \cdot 67.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 143.147 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 187.959(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 67.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.732 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 125.25)(11.98 - 2.2)0.9811/\sin(33.7)$
 $= 43.280 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811$

= 0.981 cm²

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(125.2, 2200.0, 80.0)) \cdot (70.0 - 15.0) \cdot 0.981 \\ &= 115.133 \text{ cm}^2 \text{ Includes Bevel Area of } 28.764 \text{ cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 14.98, tr = 1.666, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.139$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 70.0, tr = 1.666, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	43 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $50.76 / 105.44 = 0.481$

Weld Size Calculations, Description: K6

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (143.1467 - 1.7323 + 2 \cdot 67.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092) \\ &= 155710.86 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv \\ &= (158.4134 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\ &= 173997.53 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (158.4134 + 0.0 + 0.9811 + (107.8048)) \cdot 1092 \\ &= 291678.84 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (158.4134 + 0.0 + 0.9811 + 0.0 + (107.8048)) \cdot 1092 \\ &= 291678.84 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 432.2143 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 35629. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 151.5661) \cdot (70.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 239174. \text{ kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 432.2143 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092 \\ &= 449712. \text{ kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (35629 + 239174) = 274803 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (35629 + 0 + 449712 + 0) = 485341 \text{ kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (35629 + 449712 + 0) = 485341 \text{ kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 274802 kgf, must exceed W = 155710 kgf	or W1 = 173997 kgf
Path 2-2 = 485340 kgf, must exceed W = 155710 kgf	or W2 = 291678 kgf
Path 3-3 = 485340 kgf, must exceed W = 155710 kgf	or W3 = 291678 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.320 in.
Actual Thickness Used in Calculation	0.590 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)} \\ &= 50.76 \cdot 3470.4998 / (2 \cdot 1092 \cdot 1.0 - 0.2 \cdot 50.76) \\ &= 81.0658 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 50.76 \cdot 45.17 / (1071 \cdot 1.0 - 0.6 \cdot 50.76) \\ &= 2.2035 \text{ mm} \end{aligned}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

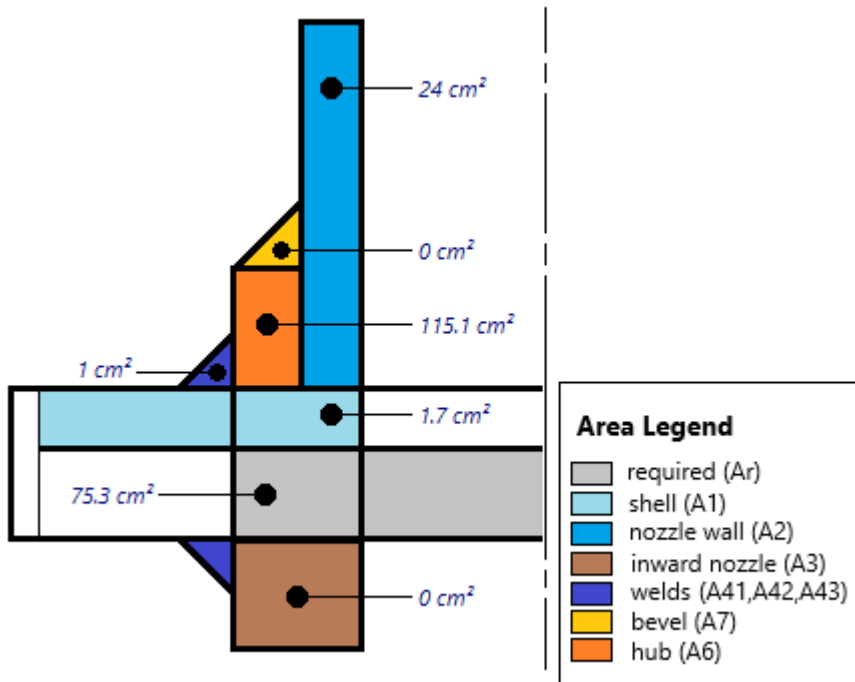
$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 55.02 / 0.5773 \end{aligned}$$

= 95.2981 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	278.3000	mm
Parallel to Vessel Wall	Rn+tn+t	139.1500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	125.2472	mm



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	75.287	NA	NA
Area in Shell	A1	1.732	NA	NA
Area in Nozzle Wall	A2	24.026	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.981	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	115.133	NA	NA
TOTAL AREA AVAILABLE	Atot	141.873	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (90.3407 \cdot 81.0658 \cdot 1.0 + 2 \cdot 67.0 \cdot 81.0658 \cdot 1.0 \cdot (1 - 0.98))$
 $= 75.287 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 187.959(1.0 \cdot 82.0 - 1.0 \cdot 81.066) - 2 \cdot 67.0$
 $(1.0 \cdot 82.0 - 1.0 \cdot 81.0658) \cdot (1 - 0.981)$
 $= 1.732 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:



$$\begin{aligned} &= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr2 \\ &= (2 \cdot 125.25) (11.98 - 2.2) \cdot 0.9811 \\ &= 24.026 \text{ cm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2 \\ &= 10.0^2 \cdot 0.9811 + (0.0)^2 \cdot 0.9811 \\ &= 0.981 \text{ cm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr2 \\ &= (2 \cdot \min(125.2, 2200.0, 80.0)) \cdot (70.0 - 15.0) \cdot 0.9811 \\ &= 115.133 \text{ cm}^2 \text{ Includes Bevel Area of } 28.764 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.2035 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 5.2035 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 14.9797 mm -> OK

Weld Size Calculations, Description: K6

Intermediate Calc. for nozzle/shell welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (75.287 - 1.7323 + 2 \cdot 67.0 \cdot 0.9811 \cdot (1.0 \cdot 82.0 - 81.0658)) \cdot 1092) \\ &= 81634.16 \text{ kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A4 - (W_i - can/0.707)^2 \cdot fr2) \cdot S_v \\ &= (139.1597 + 0.9811 - 0.0 \cdot 0.98) \cdot 1092 \\ &= 152979.83 \text{ kgf} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v \\ &= (139.1597 + 0.0 + 0.9811 + (107.8048)) \cdot 1092 \\ &= 270661.12 \text{ kgf} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (139.1597 + 0.0 + 0.9811 + 0.0 + (107.8048)) \cdot 1092 \\ &= 270661.12 \text{ kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 224.3407 \cdot 10.0 \cdot 0.49 \cdot 1071 \\ &= 18493. \text{ kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 78.6704) \cdot (70.0 - 3.0) \cdot 0.7 \cdot 1071 \\ &= 124143. \text{ kgf} \end{aligned}$$



Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 224.3407 \cdot (85.0 - 3.0) \cdot 0.74 \cdot 1092$
= 233423. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (18493 + 124143) = 142636 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (18493 + 0 + 233423 + 0) = 251916 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (18493 + 233423 + 0) = 251916 kgf

Summary of Failure Path Calculations:

Path 1-1 = 142636 kgf, must exceed W = 81634 kgf	or W1 = 152979 kgf
Path 2-2 = 251915 kgf, must exceed W = 81634 kgf	or W2 = 270661 kgf
Path 3-3 = 251915 kgf, must exceed W = 81634 kgf	or W3 = 270661 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 51 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 209.8011 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 2573.6604 mm

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: